A Controlled Study on Long Context Extension and Generalization in LLMs

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Abstract

Broad textual understanding and in-context learning require language models that utilize full document contexts. Due to the implementation challenges associated with directly training long-context models, many methods have been proposed for extending models to handle long contexts. However, owing to differences in data and model classes, it has been challenging to compare these approaches, leading to uncertainty as to how to evaluate long-context performance and whether it differs from standard evaluation. We implement a controlled protocol for extension methods with a standardized evaluation, utilizing consistent base models and extension data. Our study yields several insights into long-context behavior. First, we reaffirm the critical role of perplexity as a general-purpose performance indicator even in longer-context tasks. Second, we find that current *approximate* attention methods systematically underperform across long-context tasks. Finally, we confirm that exact fine-tuning based methods are generally effective within their extension range, whereas extrapolation remains challenging. All codebases, models, and checkpoints are made available open-source via `https://github.com/Leooyii/LCEG`, promoting transparency and facilitating further research in this critical area of AI development.

1 Introduction

The pretraining data scale of large language models (LLMs) has expanded greatly in recent years with open models trained up to 15T tokens [AI@Meta, 2024]. Implementation challenges make it difficult to fully train models with longer context windows during pretraining [Liu et al., 2023a]. Still, long-context windows are considered central, as they enable LLMs to perform tasks that require more extensive textual understanding, such as utilizing information from textbooks [Tanzer et al., 2024], summarizing novels [Kryściński et al., 2022], and engaging in many-shot learning [Bertsch et al., 2024, Li et al., 2023a].

As a trade-off, researchers have proposed *context extension*, where an LLM initially pretrained on standard sequences is adapted for significantly longer context lengths [Chen et al., 2023a, Peng et al., 2023, Han et al., 2023, bloc97, 2023]. These methods differ in the type of attention used and in post-training adaptation techniques. They vary in complexity, training requirements, and qualitatively exhibit significantly different performance profiles.

Unfortunately, there is a relatively poor understanding of the quantitative rankings of these different methodologies. Owing to the perceived challenges of evaluation, several new metrics, such as long context perplexity [Chen et al., 2023a,b, Han et al., 2023], and retrieval accuracy [Mohtashami and Jaggi, 2023, gkamradt, 2023] have been introduced [Bai et al., 2023, An et al., 2023]. However, the differences in long-context extension procedures make it hard to calibrate these metrics while controlling for other factors.

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一项关于长上下文扩展和大型语言模型泛化的控制性研究

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摘要

广泛的文本理解和上下文学习需要使用完整文档上下文的语言模型。由于直接训练长上下文模型的实施挑战，许多方法已被提出以扩展模型以处理长上下文。然而，由于数据和模型类别的差异，比较这些方法一直具有挑战性，导致不确定如何评估长上下文性能以及它是否与标准评估不同。我们实现了一个具有标准化评估的控制协议，利用一致的基座模型和扩展数据。我们的研究对长上下文行为产生了几个见解。首先，我们重申困惑度作为通用性能指标即使在更长上下文任务中的关键作用。其次，我们发现当前的近似注意力方法在长上下文任务中系统性地表现不佳。最后，我们确认基于精确微调的方法在其扩展范围内通常是有效的，而外推仍然具有挑战性。所有代码库、模型和检查点都通过
https://github.com/Leooyii/LCEG 以开源方式提供，促进透明度并促进该人工智能发展关键领域进一步研究。

1 引言

近年来，大型语言模型（LLMs）的预训练数据规模已大幅扩展，开放模型训练量高达15T tokens [AI@Meta，2024]。实施挑战使得在预训练期间难以用更长的上下文窗口完全训练模型 [Liu 等人，2023a]。尽管如此，长上下文窗口被认为是核心，因为它们使LLMs能够执行需要更广泛文本理解的任务，例如利用教科书信息 [Tanzer 等人， 2024],总结小说 [Kry.sci.nski 等人， 2022], 以及进行多示例学习 [Bertsch 等人，2024，Li 等人，2023a]。

作为权衡，研究人员提出了上下文扩展，其中在标准序列上预训练的LLM被调整为显著更长的上下文长度 [Chen 等人，2023a，Peng 等人，2023，Han 等人，2023，bloc97, 2023]。这些方法在使用的注意力类型和训练后适应技术上有所不同。它们在复杂性、训练需求和性能特征上表现出显著差异。

不幸的是，对这些不同方法论的定量排名理解相对较差。由于评估的挑战性，已经引入了几个新的指标，例如长上下文困惑度 [陈等人，2023a,b，韩等人， 2023], 以及检索准确率 [莫哈塔希米和贾吉，2023，gkamradt， 2023] [白等人，2023，安等人， 2023]。然而，长上下文扩展程序之间的差异使得在控制其他因素的情况下校准这些指标变得困难。

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In this work, we implement a controlled protocol for context extension. The aim is to compare context extension while removing spurious factors that impact LLM ability.

Modeling: We standardize on the same base model for all experiments. Different base models behave significantly differently, making it challenging to draw general conclusions. For instance, past work evaluates LM-Infinite [Han et al., 2023] on LongBench [Bai et al., 2023] using different base models [Xiao et al., 2024, Lu et al., 2024].

Extensions: We implement a range of context extension methods within the same framework. We use a standardized recipe to eliminate potential gains from tailored hyperparameters. We additionally fix the post-training data for each method, utilizing an identical and open-sourced training corpus [Fu et al., 2024, Chen et al., 2023b].

Metrics: We look at both intrinsic metrics, such as perplexity, and extrinsic properties, such as downstream task performance [Hsieh et al., 2024, gkamradt, 2023, Bai et al., 2023]. We consider metrics within the extension length as well as an extrapolation to longer contexts.

Our study identifies several takeaways for future research. First, contrary to some of the arguments for needing new metrics, our findings indicate a strong correlation between perplexity and downstream task performance for exact fine-tuned methods in controlled studies. While some approximate attention methods are unbalanced between perplexity and downstream task performance, there is a strong correlation between the two for most extension tasks.

Second, we find relatively poor results for approximate attention methods. While they can handle longer length contexts, there generally is a trade-off in terms of accuracy for most of our benchmarks. Exact frozen methods also tend to degrade model performance, showing high sensitivity to hyperparameters and often failing with a general training recipe.

Finally, continual fine-tuning with exact attention generally works well, particularly within the extended context length. Specifically, Dynamic NTK [emozilla, 2023] works best among these methods. Extrapolation to longer lengths remains a challenging problem. Our training code, models, and checkpoints will be open-sourced to support further research in this area.

2 Related Work

Long Context Methods We divide extension methods into three broad classes: exact attention, approximate attention, and context compression. Exact attention methods augment the parameterization of attention. Position interpolation (PI) [Chen et al., 2023a], NTK-aware [bloc97, 2023], Dynamic NTK [emozilla, 2023], YaRN [Peng et al., 2023], and CLEX [Chen et al., 2024], all based on RoPE [Su et al., 2021], design position embeddings for length extension. These methods may be applied with fine-tuning or to frozen models. Other exact attention methods focus on training-time improvements, such as contrastive training [Tworkowski et al., 2023] . Approximate attention methods introduce structured attention approximations that minimize the computational cost of length extension. Chen et al. [2023b] introduced the use of LoRA [Hu et al., 2021] and a specialized local attention mechanism to reduce further the computational overhead of further fine-tuning with long context. Other approaches break the text into chunks and utilize a well-designed "chunk representation" to retrieve relevant chunks for attention [Mohtashami and Jaggi, 2023, Xiao et al., 2024, Lu et al., 2024]. LM-Infinite and StreamLLM [Han et al., 2023, Xiao et al., 2023] retain only a few tokens from the beginning of the text and a local window to keep the attention window within the pretrained length. Xu et al. [2024] focuses on using retrievers to retrieve relevant blocks from long documents. Finally, context compression methods, which we do not explore in this work, reduce length extension to length compression via a summarization step [Jiang et al., 2023, Li et al., 2023b].

Long Context Evaluation Benchmarks The Long Range Arena (LRA) [Tay et al., 2020] is an early efforts in evaluating the proficiency of processing long contexts in different modalities. Since then, a growing number of benchmarks have emerged, including LongBench [Bai et al., 2023], LEval [An et al., 2023], and LooGLE [Li et al., 2023c]. These benchmarks are a mixture of diverse downstream tasks explicitly tailored to assess the capabilities of LLMs in understanding and generating lengthy contexts. Among these benchmarks, LongBench stands out for its inclusion of diverse sequences with varying lengths, distributions, patterns, languages, and domains, enabling a comprehensive, nuanced evaluation. In addition to evaluating LLMs' performance on downstream NLP tasks, there is

在这项工作中，我们实现了一种受控的上下文扩展协议。目的是在去除影响大型语言模型能力的虚假因素的同时比较上下文扩展。

建模: 我们对所有实验使用相同的基座模型进行标准化。不同的基座模型表现差异显著，使得难以得出普遍结论。例如，以往工作使用不同的基座模型[Xiao 等人，2024，Lu 等人，2024]在 LongBench [Bai 等人， 2023] 上评估 LM-Infinite [Han 等人， 2023] 。

扩展: 我们在同一个框架内实现了一系列上下文扩展方法。我们使用标准化的配方来消除来自定制超参数的潜在收益。此外，我们为每种方法固定了后训练数据，利用一个相同且开源的训练语料库 [Fu 等人，2024，Chen 等人，2023b]。

指标: 我们关注内在指标（如困惑度）和外在属性（如下游任务性能 [Hsieh 等人，2024，gkamradt，2023，Bai 等人， 2023]）。我们考虑扩展长度内的指标，以及向更长上下文的推论。

我们的研究为未来研究指出了几个要点。首先，与一些认为需要新指标的论点相反，我们的研究结果表明，在控制研究中，精确微调方法的困惑度与下游任务性能之间存在很强的相关性。虽然一些近似注意力方法在困惑度和下游任务性能之间不平衡，但对于大多数扩展任务，两者之间具有很强的相关性。

其次，我们发现近似注意力方法的结果相对较差。虽然它们可以处理更长的上下文长度，但就我们大多数基准测试而言，在准确率方面通常存在权衡。精确冻结方法也倾向于降低模型性能，显示出对超参数的高敏感性，并且通常无法使用通用的训练配方。

最后，使用精确注意力的持续微调通常效果很好，尤其是在扩展上下文长度范围内。具体来说，Dynamic NTK [emozilla, 2023] 在这些方法中表现最佳。扩展到更长的长度仍然是一个具有挑战性的问题。我们的训练代码、模型和检查点将开源，以支持该领域的进一步研究。

2 相关工作

长上下文方法 我们将扩展方法分为三大类：精确注意力、近似注意力和上下文压缩。精确注意力方法增强了注意力的参数化。位置插值（PI） [陈等人，2023a], NTK感知 [bloc97, 2023], Dynamic NTK [emozilla， 2023], YaRN [彭等人， 2023], 以及CLEX [陈等人， 2024], 都基于RoPE [苏等人， 2021], 设计用于长度扩展的位置嵌入。这些方法可以与微调一起使用或应用于冻结的模型。其他精确注意力方法专注于训练时的改进，例如对比训练 [Tworkowski等人， 2023] 。近似注意力方法引入了结构化的注意力近似，以最小化长度扩展的计算成本。陈等人 [2023b] 引入了LoRA [胡等人， 2021] 以及一种专门化的局部注意力机制，以进一步减少使用长上下文进行微调的计算开销。其他方法将文本分割成块，并利用精心设计的“块表示”来检索用于注意力的相关块 [莫哈塔希米和贾吉，2023，Xiao等人，2024，Lu等人， 2024]。LM-Infinite和StreamLLM [韩等人，2023，Xiao等人， 2023] 仅保留文本开头的一小部分标记和一个局部窗口，以保持注意力窗口在预训练长度内。Xu等人 [2024] 专注于使用检索器从长文档中检索相关块。最后，上下文压缩方法，我们在这项工作中不进行探索，通过摘要步骤将长度扩展转换为长度压缩 [姜等人，2023，李等人，2023b]。

长上下文评估基准 长程竞技场 (LRA) [Tay 等人， 2020] 是一个早期尝试评估不同模态中处理长上下文能力的努力。自那时起，越来越多的基准测试涌现出来，包括 LongBench [Bai 等人， 2023], LEval [An 等人， 2023], 和 LooGLE [Li 等人，2023c]。这些基准测试是多种下游任务的混合，这些任务明确设计用于评估大型语言模型在理解和生成长上下文方面的能力。在这些基准测试中，LongBench 因其包含不同长度、分布、模式、语言和领域的多样化序列而脱颖而出，能够进行全面且细致的评估。除了评估大型语言模型在下游自然语言处理任务上的性能外，还有

another line of benchmarks that specifically focuses on assessing particular aspects of long context processing ability [Liu et al., 2023b, Hsieh et al., 2024]. For instance, Mohtashami and Jaggi [2023] propose the passkey retrieval task to challenge a language model to accurately locate and retrieve a simple passkey (a five-digit random number) in a long context sequence. Similarly, the Needle in a Haystack [gkamradt, 2023] test requires the model to accurately recite the information from a specified sentence(the "needle") from a long document. However, most existing works mainly focus on evaluating mainstream commercial models (e.g. GPT-4 and Claude), open-source base models, or just perform individual evaluations of a few long context methods. There is a lack of comprehensive, yet controlled evaluation on long-context extension techniques themselves.

3 Context Extension Methods

3.1 Background: Attention and RoPE

The bottleneck in long context modeling in Transformers is attention. Attention is defined over C embeddings $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_C]^\top \in \mathbb{R}^{C\times d}$ where d is the model dimension. Learned weight matrices $\mathbf{W}_v \in \mathbb{R}^{d\times d_k}$, $\mathbf{W}_q \in \mathbb{R}^{d\times d_k}$, and $\mathbf{W}_k \in \mathbb{R}^{d\times d_k}$ are used to transform these inputs where d_k is the projected hidden dimension. The attention mechanism itself computes the attention matrix and applies it to produce a weighted sum of the value vectors:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathbf{AV} = \text{softmax}\left(\frac{\mathbf{QK}^\top}{\sqrt{d_k}}\right)\mathbf{V}. \tag{1}$$

Basic attention was originally defined with: $\mathbf{Q} = \mathbf{XW}_q, \mathbf{K} = \mathbf{XW}_k, \mathbf{V} = \mathbf{XW}_v$. However, this approach does not directly encode the relative position of keys and values.

Rotary Position Embeddings (RoPE) [Su et al., 2024] encode positional information by applying a phase rotation to each element of the embedding vectors. Formally, we define a transformation $\mathbf{f}$:

$$\mathbf{f}_{\mathbf{W}}(\mathbf{x}_i, \boldsymbol{\theta}) = \mathbf{R}(\boldsymbol{\theta}, i)\mathbf{W}^\top \mathbf{x}_i \tag{2}$$

Here $\mathbf{x}_i \in \mathbb{R}^{d_k}$ is an embedding for position i, $\mathbf{W}$ is a projection matrix, and $\boldsymbol{\theta} \in \mathbb{R}^{d_k/2}$ is a frequency basis. The function is defined based on the rotary position matrix:

$$\mathbf{R}(\boldsymbol{\theta}, i) = \begin{pmatrix} \cos i\theta_1 & -\sin i\theta_1 & \cdots & 0 & 0 \\ \sin i\theta_1 & \cos i\theta_1 & \cdots & 0 & 0 \\ \vdots & & & & \\ 0 & 0 & \cdots & \cos i\theta_{\frac{d_k}{2}} & -\sin i\theta_{\frac{d_k}{2}} \\ 0 & 0 & \cdots & \sin i\theta_{\frac{d_k}{2}} & \cos i\theta_{\frac{d_k}{2}} \end{pmatrix} \tag{3}$$

Due to the arrangement of frequencies, this matrix has the property that $\mathbf{R}(\boldsymbol{\theta}, n-m) = \mathbf{R}(\boldsymbol{\theta}, m)^\top \mathbf{R}(\boldsymbol{\theta}, n)$ by Ptolemy's identity. We redefine the query-key product between two positions m and n as,

$$\begin{aligned} \mathbf{q}_m^\top \mathbf{k}_n &= \mathbf{f}_{\mathbf{W}_q}(\mathbf{x}_m, \boldsymbol{\theta})^\top \mathbf{f}_{\mathbf{W}_k}(\mathbf{x}_n, \boldsymbol{\theta}) && (4) \\ &= \left(\mathbf{R}(\boldsymbol{\theta}, m)\mathbf{W}_q^\top \mathbf{x}_m\right)^\top \left(\mathbf{R}(\boldsymbol{\theta}, n)\mathbf{W}_k^\top \mathbf{x}_n\right) && (5) \\ &= \mathbf{x}_m^\top \mathbf{W}_q \mathbf{R}(\boldsymbol{\theta}, n-m)\mathbf{W}_k^\top \mathbf{x}_n && (6) \end{aligned}$$

In this way, the relative positional information $n - m$ is implicitly injected into the query and key product, thus the attention score.

The standard RoPE transformation, $f_{\mathbf{W}}(x_i, \boldsymbol{\theta})$, sets the components $\theta_j = b^{-\frac{2j}{d_k}}$ with base $b = 10000$.

另一组专门针对长上下文处理能力特定方面的评估基准 [Liu 等人，2023b，谢等人， 2024]。例如，莫哈塔希米和贾吉 [2023]提出了密钥检索任务，挑战语言模型在长上下文序列中准确定位和检索一个简单密钥（一个五位数随机数）。类似地，大海捞针 [gkamradt， 2023] 测试要求模型从长文档中准确复述指定句子的信息（即“针”）。然而，大多数现有工作主要集中于评估主流商业模型（例如 GPT-4 和 Claude）、开源基础模型，或者只是对少数长上下文方法进行单独评估。对于长上下文扩展技术本身，缺乏全面且受控的评估。

3 上下文扩展方法

3.1 背景：注意力和RoPE

Transformer中长上下文建模的瓶颈在于注意力。注意力是在 C 嵌入 $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_C]^\top \in \mathbb{R}^{C\times d}$ 上定义的，其中 d 是模型维度。学习到的权重矩阵 $\mathbf{W}_v \in \mathbb{R}^{d\times d_k}$、$\mathbf{W}_q \in \mathbb{R}^{d\times d_k}$ 和 $\mathbf{W}_k \in \mathbb{R}^{d\times d_k}$ 用于转换这些输入，其中d_k 是投影的隐藏维度。注意力机制本身计算注意力矩阵并将其应用于产生值向量的加权和：

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathbf{AV} = \text{softmax}\left(\frac{\mathbf{QK}^\top}{\sqrt{d_k}}\right)\mathbf{V}. \tag{1}$$

基本注意力最初定义为： $\mathbf{Q} = \mathbf{XW}_q$ $\mathbf{K} = \mathbf{XW}_k$ $\mathbf{V} = \mathbf{XW}_v$,。然而，这种方法没有直接编码键和值之间的相对位置。

旋转位置编码（RoPE） [Su等人， 2024] 通过将相位旋转应用于嵌入向量的每个元素来编码位置信息。形式上，我们定义一个变换 f：

$$\mathbf{f}_{\mathbf{W}}(\mathbf{x}_i, \boldsymbol{\theta}) = \mathbf{R}(\boldsymbol{\theta}, i)\mathbf{W}^\top \mathbf{x}_i \tag{2}$$

这里 $\mathbf{x}_i \in \mathbb{R}^{d_k}$ 是一个用于位置 i 的嵌入，$\mathbf{W}$ 是一个投影矩阵，而 $\boldsymbol{\theta} \in \mathbb{R}^{d_k/2}$ 是一个频率基。该函数基于旋转位置矩阵定义：

$$\mathbf{R}(\boldsymbol{\theta}, i) = \begin{pmatrix} \cos i\theta_1 & -\sin i\theta_1 & \cdots & 0 & 0 \\ \sin i\theta_1 & \cos i\theta_1 & \cdots & 0 & 0 \\ \vdots & & & & \\ 0 & 0 & \cdots & \cos i\theta_{\frac{d_k}{2}} & -\sin i\theta_{\frac{d_k}{2}} \\ 0 & 0 & \cdots & \sin i\theta_{\frac{d_k}{2}} & \cos i\theta_{\frac{d_k}{2}} \end{pmatrix} \tag{3}$$

由于频率的排列方式，该矩阵具有这样的性质： $\mathbf{R}(\boldsymbol{\theta}, n-m)$ $=\mathbf{R}(\boldsymbol{\theta}, m)^\top \mathbf{R}(\boldsymbol{\theta}, n)$ 通过托勒密定理。我们重新定义两个位置 m 和 n 之间的查询-键积为，

$$\begin{aligned} \mathbf{q}_m^\top \mathbf{k}_n &= \mathbf{f}_{\mathbf{W}_q}(\mathbf{x}_m, \boldsymbol{\theta})^\top \mathbf{f}_{\mathbf{W}_k}(\mathbf{x}_n, \boldsymbol{\theta}) && (4) \\ &= \left(\mathbf{R}(\boldsymbol{\theta}, m)\mathbf{W}_q^\top \mathbf{x}_m\right)^\top \left(\mathbf{R}(\boldsymbol{\theta}, n)\mathbf{W}_k^\top \mathbf{x}_n\right) && (5) \\ &= \mathbf{x}_m^\top \mathbf{W}_q \mathbf{R}(\boldsymbol{\theta}, n-m)\mathbf{W}_k^\top \mathbf{x}_n && (6) \end{aligned}$$

通过这种方式，相对位置信息 $n - m$ 被隐式地注入到查询和键积中，从而得到注意力分数。

标准RoPE变换， $f_{\mathbf{W}}(x_i, \boldsymbol{\theta})$，将组件 $\theta_j = b^{-\frac{2j}{d_k}}$ 设置为以 $b = 10000$为基础。

3.2 Adjusting the Frequency of RoPE for Long Context Extension

We consider four methods for performing length extension on RoPE embeddings: Position Interpolation (PI) [Chen et al., 2023a], NTK-RoPE [emozilla, 2023], YaRN [Peng et al., 2023] and CLEX [Chen et al., 2024]. In this section our goal is to extend a method trained to extend position embeddings for context length C to length $C' >> C$. The methods in this section perform this extension by scaling the frequencies with the *base scaling vector* $\boldsymbol{\alpha} \in \mathbb{R}^{\frac{d_k}{2}}$:

$$\mathbf{f}_{\mathbf{W}}(x_i) = \mathbf{f}(x_i, \boldsymbol{\alpha} \odot \boldsymbol{\theta}). \tag{7}$$

Linear Position Interpolation (PI) decreases the frequencies of the basis functions so that more tokens fit within each period. PI is implemented by setting the components of the base scaling vector to

$$\alpha_j^{\text{PI}} = \frac{C}{C'} = \frac{1}{t}. \tag{8}$$

where $t = \frac{C'}{C}$ is target length ratio. PI has been integrated into many open-source models such as LLaMA2-7B-32K [Together.AI, 2023], Vicuna-7B-v1.5 [Chiang et al., 2023], and LongAlpaca [Chen et al., 2023b].

Neural Tangent Kernel Interpolation RoPE (NTK-RoPE) builds on linear position interpolation by introducing a per-dimension scaling factor. Inspired by findings from the NTK literature that show that high-frequency features are difficult for MLPs to learn, NTK-RoPE preserves high-frequency features while extending the period of low-frequency features. This is accomplished via a dimension-dependent base scaling vector α:

$$\alpha_j^{\text{NTK-RoPE}} = \kappa^{-\frac{2j}{d_k}}, \tag{9}$$

where $\kappa = (t)^{\frac{d_k}{d_k-2}}$ so that the lowest frequency is scaled to match PI and the highest frequency remains the same as in RoPE.

An extension to this approach, Dynamic NTK-RoPE suggests that instead of fixing scaling based on a set ratio s for all examples during inference, the formula should adapt to the current context length for a specific example. We followed the set up of Fu et al. [2024] for Dynamic NTK-RoPE. More details can be found in the Appendix 9.2.

YaRN, another RoPE extension method, uses "NTK-by-parts" interpolation strategies across different dimensions of the embedding space and introduces a temperature factor to adjust the attention distribution for long inputs.

$$\alpha_j^{\text{YaRN}} = ((1-\gamma_j)\frac{1}{t} + \gamma_j)/\sqrt{T} \tag{10}$$

We use a ramp vector $\boldsymbol{\gamma}$ to determine the interpolation between the $\frac{1}{t}$ and the original frequency base. The interpolation gating is set based on the frequency for the dimension j.

$$\gamma_j = \begin{cases} 0, & \text{if } \theta_j < p, \\ 1, & \text{if } \theta_j > q, \\ \frac{\theta_j - p}{q-p}, & \text{otherwise.} \end{cases} \tag{11}$$

The values of p, q, T can be tuned as needed.

Other methods such as **CLEX** Chen et al. [2024] models the scaling vectors as a dynamical system, intending to learn target-length dependent scaling vectors.

3.3 Approximate Attention

An alternative approach is to modify the attention function itself. Instead of exactly computing each longer attention, these methods select a subset of positions to attend to. We consider four well-known methods based on three different attention mechanisms: sparse attention, sliding window attention, and retrieval attention.

3.2 调整 RoPE 的频率以实现长上下文扩展

我们考虑了四种用于对RoPE嵌入进行长度扩展的方法：位置插值（PI） [陈等人，2023a], 神经切线核插值RoPE [emozilla， 2023], YaRN [彭等人， 2023] 和CLEX [陈等人， 2024]。在本节中，我们的目标是将一种训练用于扩展位置嵌入以适应上下文长度 C 的方法扩展到长度 $C' >> C$。本节中的方法通过使用<样式id='11'>基础缩放向量</样式> $\boldsymbol{\alpha} \in \mathbb{R}^{\frac{d_k}{2}}$ 来缩放频率以执行此扩展：

$$\mathbf{f}_{\mathbf{W}}(x_i) = \mathbf{f}(x_i, \boldsymbol{\alpha} \odot \boldsymbol{\theta}). \tag{7}$$

线性位置插值（PI） 通过降低基函数的频率，使得更多 token 融合在每个周期内。PI 通过将基频缩放向量的分量设置为

$$\alpha_j^{\text{PI}} = \frac{C}{C'} = \frac{1}{t}. \tag{8}$$

其中 $t = \frac{C'}{C}$ 是目标长度比。PI 已集成到许多开源模型中，例如 LLaMA2-7B-32K [Together.AI, 2023], Vicuna-7B-v1.5 [Chiang 等人， 2023]，以及 LongAlpaca [Chen 等人，2023b]。

神经切线核插值 RoPE (NTK-RoPE) 在线性位置插值的基础上引入了每个维度的缩放因子。受 NTK 文献中关于高频特征对 MLP 学习困难的发现的启发，NTK-RoPE 在保留高频特征的同时扩展了低频特征的周期。这是通过一个维度相关的基频缩放向量α实现的：

$$\alpha_j^{\text{NTK-RoPE}} = \kappa^{-\frac{2j}{d_k}}, \tag{9}$$

在 $\kappa = (t)^{\frac{d_k}{d_k-2}}$ 处，以便将最低频率缩放到匹配 PI，并且最高频率保持与 RoPE 相同。

这种方法的扩展，Dynamic NTK-RoPE 建议在推理过程中，不是基于固定比例 s 为所有示例设置缩放，而是公式应该适应特定示例的当前上下文长度。我们遵循了 Fu 等人 [2024] 为 Dynamic NTK-RoPE 设置的方案。更多细节可以在附录 9.2 中找到。

YaRN，另一种 RoPE 扩展方法，在不同的嵌入空间维度上使用“NTK-by-parts”插值策略，并引入温度因子来调整长输入的注意力分布。

$$\alpha_j^{\text{YaRN}} = ((1-\gamma_j)\frac{1}{t} + \gamma_j)/\sqrt{T} \tag{10}$$

我们使用一个斜坡向量 $\boldsymbol{\gamma}$ 来确定 $\frac{1}{t}$ 与原始频率基础之间的插值。插值门控基于维度 j 的频率设置。

$$\gamma_j = \begin{cases} 0, & \text{if } \theta_j < p, \\ 1, & \text{if } \theta_j > q, \\ \frac{\theta_j - p}{q-p}, & \text{otherwise.} \end{cases} \tag{11}$$

p, q, T 的值可以根据需要调整。

其他方法，例如 **CLEX** Chen 等人 [2024] 将缩放向量建模为一个动力系统，旨在学习目标长度相关的缩放向量。

3.3 近似注意力

另一种方法是修改注意力函数本身。这些方法不是精确计算每个较长的注意力，而是选择一部分位置来关注。我们考虑了基于三种不同注意力机制的四种著名方法：稀疏注意力、滑动窗口注意力和检索注意力。

LongLoRA [Chen et al., 2023b] avoids computing attention ranges over C' by only computing the block-diagonl part of attention. Formally, given a sequence length of C', LongLoRA divides it into M blocks of size B, resulting in a sparse attention matrix $\mathbf{A} \in \mathbb{R}^{C' \times C'}$ with a block-diagonal structure:

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{A}_M \end{bmatrix} \tag{12}$$

where $\mathbf{A}_i \in \mathbb{R}^{B \times B}$ is the attention matrix for the i-th block. In addition, they shift the blocks for half of the heads enabling the information flow between groups via shifting. Notably, while they employ local attention during the fine-tuning phase, full attention is still adopted during the inference stage.

Landmark Attention [Mohtashami and Jaggi, 2023] addresses the challenge of attending over long sequences by breaking the input sequence into chunks and using trainable "landmark" tokens to summarize these chunks. The attention process is carried out in two stages. Given a sequence of C' embeddings, divided into M chunks, each of length B, the first step is to compute global attention between the query vectors $\mathbf{Q} \in \mathbb{R}^{C' \times d_k}$ (corresponding to all input embedding) and the landmark vectors $\mathbf{L} \in \mathbb{R}^{M \times d_k}$ (which represent the chunks). From this global attention, a set of n-most attended-to chunks is selected for further processing. Next, a local attention mechanism is applied within each of the selected chunks. For the n-th selected chunk, the key matrix for the chunk is denoted as $\mathbf{K}_n \in \mathbb{R}^{B \times d_k}$ and $\mathbf{Q}_n \in \mathbb{R}^{B \times d_k}$. The attention matrices are then computed as follows:

$$\mathbf{A}^1 = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{L}^T}{\sqrt{d_k}}\right) \in \mathbb{R}^{C' \times M}, \tag{13}$$

$$\mathbf{A}^{2,n} = \text{softmax}\left(\frac{\mathbf{Q}_n\mathbf{K}_n^T}{\sqrt{d_k}}\right) \in \mathbb{R}^{B \times B}, \tag{14}$$

The final attention for each embedding is a combination of these two attention. This method efficiently scales attention mechanisms for long sequences by focusing on landmark tokens that summarize large parts of the sequence, followed by local attention within the relevant chunks.

LM-Infinite [Han et al., 2023] (a.k.a., Sliding Window Attention) maintains a sliding local window of size M along with a fixed global memory of G positions at the starting point of the given embedding. Given C' embeddings, attention is computed over the M embeddings in its local window and G embeddings in global memory. LM-Infinite replaces relative positional information $n - m$ with $\min(n - m, C)$ while computing the query and key product in Eq 4. Altogether, LM-Infinite reduces the complexity from $\mathcal{O}((C')^2)$ to $\mathcal{O}(C'(M + G))$ without the need to scale positional encoding.

Self-Extend [Jin et al., 2024] maps the unseen positions in extended context length C' to positions within the pretraining context length C to avoid training. For each embeddings, Self-Extend chooses closest M embeddings and any embeddings beyond are divided into multiple groups. Each group contains N embeddings. When performing query-key product between two positions m and n in Equation 4, the relative positional information $n - m$ is replaced by r which is computed by scaling $n - m$ w.r.t M and N:

$$r = \begin{cases} n - m, & n - m \le M, \\ M + \left\lfloor \frac{n-m}{N} \right\rfloor - \left\lfloor \frac{M}{N} \right\rfloor, & n - m > M. \end{cases} \tag{15}$$

where $\lfloor \cdot \rfloor$ denotes the floor division. The maximum extended context length C' is $(C - M) \cdot N + M$.

4 Long-Context Extension Protocol

Base Model All models start from an identical LLaMA2-7B base checkpoint [Touvron et al., 2023]. This approach removes potential biases from the checkpoint basis. Note that to study the influence of base model, particularly the size of base model, we also use Phi-2-base checkpoint [Javaheripi et al., 2023] for context extension, to verify whether the trends and analyses we observed are consistent across different base models, thereby avoiding potential over-generalization. The results of Phi-2-base are reported in Appendix 9.1.

LongLoRA [陈等人，2023b] 通过仅计算注意力的块对角部分来避免计算注意力范围， C' 。形式上，给定序列长度为 C'，LongLoRA将其分成 M 个大小为 B的块，从而得到一个具有块对角结构的稀疏注意力矩阵 $\mathbf{A} \in \mathbb{R}^{C' \times C'}$：

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{A}_M \end{bmatrix} \tag{12}$$

其中 $\mathbf{A}_i \in \mathbb{R}^{B \times B}$ 是 i 块的注意力矩阵。此外，他们通过移动块来使信息在头之间流动，移动了头的一半。值得注意的是，在微调阶段他们使用局部注意力，但在推理阶段仍然采用全注意力。

地标注意力 [莫哈塔希米和贾吉， 2023] 通过将输入序列分割成块并使用可训练的“地标”标记来总结这些块，解决了在长序列中注意力的挑战。注意力过程分为两个阶段。给定一个由 C' 嵌入组成的序列，将其分割成 M 块，每个块的长度为 B，第一步是计算查询向量 $\mathbf{Q} \in \mathbb{R}^{C' \times d_k}$ （对应于所有输入嵌入）和地标向量 $\mathbf{L} \in \mathbb{R}^{M \times d_k}$ （代表块）之间的全局注意力。从这个全局注意力中，选择一组n最受关注的块进行进一步处理。接下来，在每个选定的块内应用局部注意力机制。对于第n个选定的块，块的键矩阵表示为 $\mathbf{K}_n \in \mathbb{R}^{B \times d_k}$ 和 $\mathbf{Q}_n \in \mathbb{R}^{B \times d_k}$。注意力矩阵的计算方法如下：

$$\mathbf{A}^1 = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{L}^T}{\sqrt{d_k}}\right) \in \mathbb{R}^{C' \times M}, \tag{13}$$

$$\mathbf{A}^{2,n} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}_n^T}{\sqrt{d_k}}\right) \in \mathbb{R}^{B \times B}, \tag{14}$$

最终的注意力机制对于每个嵌入是一个由这两种注意力机制组合而成的结果。这种方法通过关注总结序列大部分内容的landmark标记，有效地扩展了注意力机制，以处理长序列，然后在其相关块内进行局部注意力。

LM-Infinite [韩等人， 2023](又名，滑动窗口注意力)维护一个大小为 M 的滑动局部窗口，以及一个在给定嵌入起始点处固定大小的 G 位置全局内存。给定 C' 嵌入，注意力机制在其局部窗口内的 M 嵌入和全局内存内的 G嵌入之间进行计算。LM-Infinite在计算公式4中的查询和键产品时，用min $(n-m, C)$ 替换了相对位置信息$n-m$ 。总而言之，LM-Infinite将复杂度从 $\mathcal{O}((C')_2)$降低到 $\mathcal{O}(C'(M+G))$，而无需扩展位置编码。

Self-Extend [Jin 等人， 2024] 将扩展上下文长度 C' 中的未见过位置映射到预训练上下文长度 C 中的位置，以避免训练。对于每个嵌入，Self-Extend 选择最接近的 M 嵌入，而任何超出范围的嵌入则被分成多个组。每个组包含 N 个嵌入。当在公式 4 中对两个位置 m 和 n 进行查询-键积时，相对位置信息 $n-m$ 被替换为 r，该值通过将 $n-m$ 相对于 M 和 N 进行缩放计算得到:

$$r = \begin{cases} n - m, & n - m \le M, \\ M + \left\lfloor \frac{n-m}{N} \right\rfloor - \left\lfloor \frac{M}{N} \right\rfloor, & n - m > M. \end{cases} \tag{15}$$

其中 $\lfloor \cdot \rfloor$ 表示向下取整除法。最大扩展上下文长度 C' 是 $(C-M) \cdot N + M$。

4 Long-ContextExtension 协议

基础模型 所有模型都从一个相同的 LLaMA2-7B 基础检查点 [Touvron 等人， 2023]开始。这种方法消除了检查点基础上的潜在偏差。请注意，为了研究基础模型的影响，特别是基础模型的大小，我们还使用 Phi-2-base 检查点 [Javaheripi 等人,2023]进行上下文扩展，以验证我们观察到的趋势和分析是否在不同的基础模型上一致，从而避免潜在的过度泛化。Phi-2-base 的结果报告在附录9.1中。

Fine-Tuning We sample 1B tokens from a long-context data mixture from Fu et al. [2024] to achieve state-of-the-art context extension performance. The details of the data are reported in Appendix 9.3. We focus on extending the context window from 4k to 32k, as most downstream tasks require contexts under 32k.

We maintain a fixed training recipe to ensure consistency across all models [Chen et al., 2023b]. We follow existing practices by keeping an exponential moving average (EMA) of model weights with a constant decay. Most training hyperparameters are based on [Fu et al., 2024], with the learning rate set to 2×10^{-5}. We use a linear learning rate warm-up and zero weight decay, utilizing 8 NVIDIA A100 GPUs.

For LongLora, we fine-tune the weights of the LoRA adapter with trainable embedding and normalization, then merge these trainable weights with the LLaMA2 base model for evaluation. For Landmark Attention, the training context length is set to 512, with a block size of 64. For CLEX, we set the max scale factor to 32 and use the SiLU activation function.

We reuse the original scale factor to maintain consistency for NTK, YaRN, and Position Interpolation methods. However, this base factor significantly degrades continual fine-tuned models, particularly causing performance deterioration in shorter sequences. Therefore, we conduct a grid search to determine a better scale factor for different input lengths for NTK-RoPE method. Based on our findings, we follow and improve upon Fu et al. [2024] to set the scale factor for NTK-RoPE method. The scale factor and its relationship with perplexity are reported in the Appendix 9.5. Please refer to the Appendix 9.2 for detailed hyperparameter setups.

Metrics We consider two sets of intrinsic metrics. The first is based on *perplexity*. We use the book corpus PG19 [Rae et al., 2019] and the Proof-pile dataset [Azerbayev et al., 2023] to evaluate the long sequence language modeling performances. Following Press et al. [2022], all perplexity evaluations are calculated using a sliding window with a window size of 256.

The second is based on *retrieval*. We focus on the needle in the haystack task [gkamradt, 2023](NIAH). NIAH involves identifying a specific, relevant piece of information (the "needle") within a large set of irrelevant data (the "haystack"). This task is commonly used to test the precision and recall capabilities of LLMs in scenarios where the relevant data is sparse and surrounded by a significant amount of noise. Additionally, we evaluate with RULER [Hsieh et al., 2024]. RULER enhances the standard NIAH test by incorporating variations with different types and quantities of needles. Additionally, it introduces new task categories, such as multi-hop tracing and aggregation, to evaluate behaviors beyond simple context-based searching.

For extrinsic metrics, we consider a collection of tasks. LongBench [Bai et al., 2023] is a family of bilingual, multitask evaluations for long-context understanding widely used in measuring the long-context abilities of LLMs [Jin et al., 2024, Xiao et al., 2024, Lu et al., 2024]. LongBench includes single-document question answering, multi-document QA, summarization, few-shot learning, and code completion. We follow Bai et al. [2023] to evaluate the models on 32k context window sizes by truncating the prompt from the middle when the task length exceeds a designated context window size. We also consider the ManyShots tasks, where the long-context model will be given several examples as prompts. We use the Trec News [Kontonis et al., 2024] dataset for this task.

5 Experimental Results

5.1 Result Overview

Table 1 overviews the results across both types of evaluation. The main result demonstrate that fine-tuned exact attention methods for long contexts, such as NTK-32K and YARN, consistently outperform approximate attention methods and frozen methods by a significant margin. This suggests that trading accuracy for speed in approximate attention methods can result in a loss of important reasoning capabilities, particularly for retrieval-based tasks. The performance disparity highlights the importance of exact attention mechanisms in maintaining high accuracy over extended contexts, emphasizing the need for careful consideration of attention mechanisms in model design for long-context tasks. We now consider each type of result in more detail.

微调 我们从 Fu 等人 [2024] 的长上下文数据混合中采样 1B 个标记，以实现最先进的上下文扩展性能。数据的详细信息报告在附录9.3中。我们专注于将上下文窗口从 4k 扩展到 32k，因为大多数下游任务需要上下文小于 32k。

我们维护一个固定的训练配方，以确保所有模型 [Chenet等人，2023b]的一致性。我们遵循现有实践，通过保持模型权重的指数移动平均（EMA）并使用恒定衰减来做到这一点。大多数训练超参数基于 [Fu等人， 2024], ，并将学习率设置为 2×10^{-5}。我们使用线性学习率预热和零权重衰减，利用8个NVIDIA A100 GPU。

对于LongLora，我们微调LoRA适配器的权重，包括可训练的嵌入和规范化，然后将这些可训练的权重与LLaMA2基础模型合并以进行评估。对于Landmark Attention，训练上下文长度设置为512，块大小为64。对于CLEX，我们将最大缩放因子设置为32，并使用SiLU激活函数。

我们重用原始缩放因子，以保持NTK、YaRN和位置插值方法的一致性。然而，这个基础因子会显著降低持续微调模型，特别是在较短的序列中会导致性能下降。因此，我们针对NTK-RoPE方法的不同输入长度进行网格搜索，以确定更好的缩放因子。根据我们的发现，我们遵循并改进了Fu等人 [2024]，为NTK-RoPE方法设置缩放因子。缩放因子及其与困惑度的关系在附录9.5中报告。请参考附录9.2以获取详细的超参数设置。

指标 我们考虑两组内在指标。第一组基于 困惑度。我们使用书籍语料库 PG19 [Rae 等人， 2019] 以及 Proof-pile 数据集 [Azerbayev 等人， 2023] 来评估长序列语言建模性能。根据 Press 等人 [2022], ，所有困惑度评估都使用滑动窗口计算，窗口大小为 256。

第二组基于检索。我们关注针锋相对任务 [gkamradt， 2023](NIAH)。NIAH 涉及在大量无关数据（“干草堆”）中识别一个特定且相关的信息片段（“针”）。此任务通常用于测试大型语言模型在相关数据稀疏且被大量噪声包围的场景中的精确度和召回能力。此外，我们使用 RULER [Hsieh 等人， 2024]进行评估。RULER 通过引入不同类型和数量的针的变体来增强标准 NIAH 测试。此外，它引入了新的任务类别，如多跳推理和聚合，以评估超出简单基于上下文搜索的行为。

对于外部指标，我们考虑一组任务。LongBench [Bai 等人， 2023] 是一个用于长上下文理解的双语多任务评估系列，被广泛用于衡量大型语言模型 [Jin 等人，2024，Xiao 等人，2024，Lu 等人，2024]的长期上下文能力。LongBench包括单文档问答、多文档问答、摘要、少样本学习和代码补全。我们遵循Bai 等人 [2023]，在任务长度超过指定上下文窗口大小时，通过从中间截断提示来评估模型在32k上下文窗口大小上的表现。我们还考虑ManyShots任务，其中将向长上下文模型提供几个示例作为提示。我们使用Trec News [Kontonis 等人， 2024] 数据集来执行此任务。

5 实验结果

5.1 结果概述

表1概述了两种评估类型的结果。主要结果表明，针对长上下文的微调精确注意力方法（如NTK-32K和YaRN）在显著差距上始终优于近似注意力方法和冻结方法。这表明在近似注意力方法中用速度换取准确率可能导致重要推理能力的损失，尤其是在基于检索的任务中。性能差异突显了精确注意力机制在保持长上下文中高准确率的重要性，强调了在长上下文任务中模型设计中仔细考虑注意力机制的必要性。我们现在将更详细地考虑每种类型的结果。

Table 1: Overview of results across different extension types.

Attention Mechanisms		Model	PPL	Needle	Mshots	LongB	RULER
	Frozen	NTK-F	14.52	18.8	64.5	25.54	0.72
Exact Attention	Fine-Tuned	PI	5.85	42.1	**75.5**	33.48	57.66
		YaRN	5.85	46.7	75.0	33.45	36.95
		CLEX	5.82	71.1	74.0	33.48	52.17
		NTK-32K	**5.79**	**83.7**	71.0	**35.32**	59.42
		NTK-64K	5.93	69.1	73.0	34.30	**60.03**
Approxi. Attention	Frozen	LM-Infinite	6.71	23.9	61.5	25.84	12.34
		Self-Extend	6.11	25.8	72.0	33.62	29.50
	Fine-tuned	LongLora	9.89	20.3	55.5	23.30	3.53
		Landmark	8.13	50.9	50.0	28.19	13.56

Figure 1: Needle in a Haystack evaluation. Green squares indicates a high retrieval success rate, the white dashed line denotes the longest length examples seen at training or finetuning, and the Y-axis represents the distance to the retrieved target.

5.2 Intrinsic tasks

Perplexity Table 2 shows perplexity scores across length. We see that continuous fine-tuning methods like PI, YaRN, and LongLora effectively keep low perplexity scores within the pre-training context length. However, when the context length exceeds perplexity scores escalate once the context surpasses the pre-trained window. Only NTK and CLEX can generalize to unseen sequence length in both pretraining and continual finetuning. Additionally, we find that exact attention maintains better perplexity than LoRA, which may reduce LongLora's ability. We also note that results on both PG19 and Proof-file gave nearly consistent conclusions.

表1：不同扩展类型的结果概述。

注意力机制		模型	PPL	针	Mshots	LongB	RULER
	冻结	NTK-F	14.52	18.8	64.5	25.54	0.72
精确注意力	Fine-Tuned	PI	5.85	42.1	**75.5**	33.48	57.66
		YaRN	5.85	46.7	75.0	33.45	36.95
		CLEX	5.82	71.1	74.0	33.48	52.17
		NTK-32K	**5.79**	**83.7**	71.0	**35.32**	59.42
		NTK-64K	5.93	69.1	73.0	34.30	**60.03**
近似注意力	冻结	LM-Infinite	6.71	23.9	61.5	25.84	12.34
		Self-Extend	6.11	25.8	72.0	33.62	29.50
	Fine-tuned	LongLora	9.89	20.3	55.5	23.30	3.53
		地标	8.13	50.9	50.0	28.19	13.56

图1：大海捞针评估。绿色方块表示高检索成功率，白色虚线表示训练或微调时遇到的最长长度示例，Y轴表示检索目标距离。

5.2 内在任务

困惑度 表2显示了困惑度得分随长度变化的情况。我们看到，像PI、YaRN和LongLora这样的连续微调方法在预训练上下文长度内有效保持较低的困惑度得分。然而，当上下文长度超过预训练窗口时，困惑度得分会上升。只有NTK和CLEX在预训练和持续微调中都能泛化到未见过的序列长度。此外，我们发现精确注意力比LoRA保持更好的困惑度，这可能降低了LongLora的能力。我们还注意到，在PG19和证明文件上的结果几乎得出了一致的结论。

Table 2: Perplexity results of different methods on PG 19 and Proof-file. NTK-32K and NTK-64K refer to NTK-Dynamic, which requires finetuning on longer text. Len refers to the longest-length examples seen at training or fine-tuning. Ex refers to the exact attention. All results are produced by our experiments.

Model Details				Eval Length					
	Len	Ex	Methods	2k	4k	8k	16k	32k	64k
PG19									
Frozen	4k	✓	LLaMA2	**6.61**	**6.30**	-	-	-	-
	4k		LM-Infinite	**6.61**	**6.30**	6.25	6.45	6.71	8.49
	4k	✓	NTK-Frozen	**6.61**	**6.30**	6.82	7.94	14.52	-
	4k		Self-Extend	**6.61**	6.32	6.15	6.07	6.11	7.15
Finetuned	32k	✓	PI	6.88	6.52	6.27	6.08	5.95	-
	32k	✓	NTK-32K	6.63	6.32	**6.09**	**5.92**	**5.79**	**5.76**
	32k	✓	YaRN	6.70	6.39	6.16	6.01	5.93	-
	32k	✓	CLEX	6.85	6.62	6.14	5.93	5.82	5.79
	32k		LongLora	12.80	11.52	10.70	10.18	9.89	-
	32k		Landmark	8.15	8.14	8.14	8.11	8.13	8.15
	64k	✓	NTK-64K	6.83	6.49	6.25	6.07	5.93	5.85
Proof-file									
Frozen	4k	✓	LLaMA2	3.34	3.04	-	-	-	-
	4k		LM-Infinite	3.34	3.04	2.94	3.02	3.11	3.12
	4k	✓	NTK-Frozen	3.34	3.04	2.91	3.09	4.06	12.65
	4k		Self-Extend	3.35	3.06	2.88	2.78	2.75	2.90
Finetuned	32k	✓	PI	3.34	3.03	2.83	2.68	2.58	-
	32k	✓	NTK-32K	**3.27**	**2.98**	**2.78**	**2.64**	**2.54**	**2.48**
	32k	✓	YaRN	3.29	3.00	2.81	2.68	2.59	106.38
	32k	✓	CLEX	3.37	3.10	2.80	2.65	2.55	**2.48**
	32k		LongLora	5.97	5.10	4.58	4.27	4.13	-
	32k		Landmark	4.51	4.50	4.48	4.49	4.49	4.49
	64k	✓	NTK-64K	3.33	3.03	2.83	2.69	2.58	2.51

Needle-in-the-haystack NIAH results are shown in Figure 1. Continuous finetuning approaches such as NTK, PI, and YaRN have successfully retrieved the "needle" within the pretraining length. Yet, only the NTK and CLEX method can retrieve the needle beyond the pretraining length, aligning with the perplexity results. The performance of the Exact Attention Method generally surpasses that of the Approximate Attention Methods. LM-Infinite and Landmark Excel are only within the local window, and they struggle to retrieve the intermediate text accurately. Regarding the Dynamic NTK method, NTK-F exhibits weak generalization when not trained. When trained on the same amount of data(1B), NTK-32K outperforms NTK-64K. When trained on 2B tokens, NTK-64K demonstrated a significant performance improvement, details are in Appendix 9.4.

RULER We test all models on 13 diverse tasks from the four RULER categories [Hsieh et al., 2024]. Each model is evaluated with 500 examples for lengths of 4k, 8k, 16k, 32k, and 64k. Results are compared with the Llama2-7B baseline in Table 3. We observed a similar trend as in the NIHK task, where NTK family methods performed the best. NTK-32k maintained relatively good performance compared to other methods finetuned with a length cap of 32k. Performance of models on different length and breakdown by 13 subtasks can be found in Appendix 9.8.

5.3 Extrinsic tasks

LongBench The evaluation results of most methods on LongBench are presented in Table 4, and results on all methods are presented in Appendix 9.7. Both LM-Infinite and Landmark Attention

表2：PG 19和证明文件上不同方法的困惑度结果。NTK-32K和NTK-64K指NTK-Dynamic，需要在更长的文本上进行微调。Len指训练或微调时遇到的最长长度的示例。Ex指精确注意力。所有结果均由我们的实验产生。

Model Details				评估长度					
	Len	Ex	方法	2k	4k	8k	16k	32k	64k
PG19									
冻结	4k	✓	LLaMA2	**6.61**	**6.30**	-	-	-	-
	4k		LM-Infinite	**6.61**	**6.30**	6.25	6.45	6.71	8.49
	4k	✓	NTK-Frozen	**6.61**	**6.30**	6.82	7.94	14.52	-
	4k		Self-Extend	**6.61**	6.32	6.15	6.07	6.11	7.15
微调	32k	✓	PI	6.88	6.52	6.27	6.08	5.95	-
	32k	✓	NTK-32K	6.63	6.32	**6.09**	**5.92**	**5.79**	**5.76**
	32k	✓	YaRN	6.70	6.39	6.16	6.01	5.93	-
	32k	✓	CLEX	6.85	6.62	6.14	5.93	5.82	5.79
	32k		LongLora	12.80	11.52	10.70	10.18	9.89	-
	32k		地标	8.15	8.14	8.14	8.11	8.13	8.15
	64k	✓	NTK-64K	6.83	6.49	6.25	6.07	5.93	5.85
证明文件									
冻结	4k	✓	LLaMA2	3.34	3.04	-	-	-	-
	4k		LM-Infinite	3.34	3.04	2.94	3.02	3.11	3.12
	4k	✓	NTK-Frozen	3.34	3.04	2.91	3.09	4.06	12.65
	4k		Self-Extend	3.35	3.06	2.88	2.78	2.75	2.90
微调	32k	✓	PI	3.34	3.03	2.83	2.68	2.58	-
	32k	✓	NTK-32K	**3.27**	**2.98**	**2.78**	**2.64**	**2.54**	**2.48**
	32k	✓	YaRN	3.29	3.00	2.81	2.68	2.59	106.38
	32k	✓	CLEX	3.37	3.10	2.80	2.65	2.55	**2.48**
	32k		LongLora	5.97	5.10	4.58	4.27	4.13	-
	32k		地标	4.51	4.50	4.48	4.49	4.49	4.49
	64k	✓	NTK-64K	3.33	3.03	2.83	2.69	2.58	2.51

寻找针NIAH 结果在图 1 中显示。连续微调方法（如 NTK、PI 和 YaRN）已成功在预训练长度内检索到“针”。然而，只有 NTK 和 CLEX 方法能够在预训练长度之外检索到针，这与困惑度结果一致。精确注意力方法的性能通常优于近似注意力方法。LM-Infinite 和 Landmark Excel 仅在局部窗口内，难以准确检索中间文本。关于动态 NTK 方法，NTK-F 在未经训练时泛化能力较弱。当使用相同数量的数据（1B）进行训练时，NTK-32K 的表现优于 NTK-64K。当使用 2B token 进行训练时，NTK-64K 表现出显著的性能提升，详细信息见附录 9.4。

RULER 我们在来自 RULER 四个类别的 13 项不同任务上测试了所有模型 [谢等人， 2024]. 每个模型使用 500 个长度为 4k、8k、16k、32k 和 64k 的示例进行评估。结果与表 3 中的 Llama2-7B 基线进行了比较。我们观察到与 NIHK 任务中类似的趋势，其中 NTK 系列方法表现最佳。NTK-32k 相对于其他使用 32k 长度限制进行微调的方法，保持了相对较好的性能。不同长度下模型的表现以及按 13 个子任务分解的结果可以在附录 9.8 中找到。

5.3 外部任务

LongBench LongBench 上的大多数方法评估结果如表 4 所示，所有方法的结果在附录 9.7 中呈现。LM-Infinite 和 Landmark 注意力

Table 3: RULER evaluation. Performance of models evaluated at length from 4k to 64k. Each score is computed by averaging the accuracy of 13 tasks. Train Len refers to the longest-length examples seen at continuous finetuning.

	Models	Train Len	4k	8k	16k	32k	64k
Frozen	LLaMA2	4k	80.94	-	-	-	-
	LM-Infinite	4k	81.05	30.01	18.02	12.34	10.56
	NTK-Frozen	4k	81.14	44.45	14.79	0.72	0.91
	Self-Extend	4k	65.03	50.73	44.02	29.50	9.34
Finetuned	PI	32k	84.56	76.04	69.64	57.66	0.00
	NTK-32K	32k	86.58	**77.75**	**70.01**	59.42	46.26
	YaRN	32k	79.12	65.60	54.21	36.95	0.00
	CLEX	32k	50.18	63.93	64.35	52.17	30.61
	LongLora	32k	10.58	6.37	3.67	3.53	0.00
	Landmark	32k	22.37	17.52	16.31	13.56	14.15
	NTK-64K	64k	**86.60**	76.34	69.56	**60.03**	**49.31**

Table 4: LongBench results. N-32 and N-64 refer to NTK finetuned on 32K and 64K context lengths respectively. SE refers to Self-Extend. YN refers to YaRN. CX refers to CLEX. LLR refers to LongLora. Len refers to average length of the datasets. Train Len refers to the longest length examples seen at training or finetuning. Eval Len refers to the maximum length of the input prompt. ✓refers to whether the method is exact attention.

		Frozen			Finetuned						
	Len	Base	N-F	SE	PI	N-32	YN	CX	LLR	Land	N-64
		✓	✓		✓	✓	✓			✓	
Train Len		4k	4k	4K	32k	32k	32k	32k	32k	32k	64k
Eval Len		4k	32k	32K	32k	32k	32k	32k	32k	32k	32k
NQA	18k	21.09	3.88	23.49	23.02	23.73	19.82	24.19	12.07	12.47	**24.31**
QAP	4k	26.94	26.79	28.75	25.85	**27.50**	26.98	23.36	20.15	19.06	24.97
MQA	5k	32.42	29.82	32.66	35.10	38.22	37.11	40.83	24.50	21.86	**40.60**
HQA	9k	31.23	32.10	37.63	36.98	**41.56**	38.60	35.59	27.41	33.66	41.47
WQA	5k	25.75	22.34	30.70	29.38	**31.58**	30.63	28.24	21.46	24.94	28.62
MSQ	11k	10.55	8.84	15.73	16.80	17.41	**22.08**	17.12	11.46	11.41	18.24
GR	9k	17.32	17.87	13.15	25.61	**28.27**	20.98	24.68	24.05	17.20	24.37
QSM	11k	21.28	15.35	20.20	21.19	21.52	20.66	21.55	17.66	18.83	**21.65**
MWS	2k	3.44	9.30	1.50	10.55	22.13	8.91	16.96	21.19	19.43	**25.02**
TRE	5k	66.00	67.50	69.00	**71.00**	69.00	69.00	67.50	50.00	49.00	69.00
TQA	8k	87.89	18.69	88.44	88.55	88.86	**89.63**	89.36	12.28	74.75	88.65
SMS	6k	41.70	32.46	43.76	**43.35**	42.21	44.25	43.02	13.45	40.38	41.59
PSC	11k	2.10	2.67	0.00	1.50	2.68	1.05	2.50	**4.57**	0.64	2.09
PSR	9k	**9.00**	3.77	4.50	4.50	4.62	3.79	8.50	3.50	2.50	6.50
LCC	1k	**68.22**	63.64	68.47	55.05	56.78	54.06	49.45	57.12	56.70	52.04
REP	4k	**61.73**	53.69	59.99	47.26	49.09	47.60	42.84	51.92	48.23	39.68
Avg.	7k	32.92	25.54	33.62	33.48	**35.32**	33.45	33.48	23.30	28.19	34.30

exhibit significant performance degradation compared to the base model. In contrast, the NTK, PI, and YaRN methods have successfully maintained their performance at 32k, demonstrating comparable results among these methods. This suggests that PI and YaRN perform similarly in downstream tasks, while the NTK family of models remains stable.

Notably, the LongLoRA method, which utilizes LoRA, also experiences a performance decline relative to the base checkpoint, LLaMA2. We argue that this may be due to the sensitivity of the training procedures for LongLoRA, and we acknowledge this in our limitation discussion section.

表3：RULER评估。在长度从4k到64k下评估模型的性能。每个分数是通过平均13个任务的准确率计算的。TrainLen指的是在连续微调中遇到的最长长度的示例。

	模型	Train Len	4k	8k	16k	32k	64k
冻结	LLaMA2	4k	80.94	-	-	-	-
	LM-Infinite	4k	81.05	30.01	18.02	12.34	10.56
	NTK-Frozen	4k	81.14	44.45	14.79	0.72	0.91
	Self-Extend	4k	65.03	50.73	44.02	29.50	9.34
微调	PI	32k	84.56	76.04	69.64	57.66	0.00
	NTK-32K	32k	86.58	**77.75**	**70.01**	59.42	46.26
	YaRN	32k	79.12	65.60	54.21	36.95	0.00
	CLEX	32k	50.18	63.93	64.35	52.17	30.61
	LongLora	32k	10.58	6.37	3.67	3.53	0.00
	地标	32k	22.37	17.52	16.31	13.56	14.15
	NTK-64K	64k	**86.60**	76.34	69.56	**60.03**	**49.31**

表4：LongBench结果。N-32和N-64分别指在32k和64k上下文长度上微调的NTK。SE指Self-Extend。YN指YaRN。CX指CLEX。LLR指LongLora。Len指数据集的平均长度。Train Len指训练或微调时遇到的最长长度示例。Eval Len指输入提示的最大长度.✓指该方法是否为精确注意力。

		冻结			微调						
	Len	Base	N-F	SE	PI	N-32	YN	CX	LLR	Land	N-64
		✓	✓		✓	✓	✓			✓	
Train Len		4k	4k	4K	32k	32k	32k	32k	32k	32k	64k
Eval Len		4k	32k	32K	32k	32k	32k	32k	32k	32k	32k
NQA	18k	21.09	3.88	23.49	23.02	23.73	19.82	24.19	12.07	12.47	**24.31**
QAP	4k	26.94	26.79	28.75	25.85	**27.50**	26.98	23.36	20.15	19.06	24.97
MQA	5k	32.42	29.82	32.66	35.10	38.22	37.11	40.83	24.50	21.86	**40.60**
HQA	9k	31.23	32.10	37.63	36.98	**41.56**	38.60	35.59	27.41	33.66	41.47
WQA	5k	25.75	22.34	30.70	29.38	**31.58**	30.63	28.24	21.46	24.94	28.62
MSQ	11k	10.55	8.84	15.73	16.80	17.41	**22.08**	17.12	11.46	11.41	18.24
GR	9k	17.32	17.87	13.15	25.61	**28.27**	20.98	24.68	24.05	17.20	24.37
QSM	11k	21.28	15.35	20.20	21.19	21.52	20.66	21.55	17.66	18.83	**21.65**
MWS	2k	3.44	9.30	1.50	10.55	22.13	8.91	16.96	21.19	19.43	**25.02**
TRE	5k	66.00	67.50	69.00	**71.00**	69.00	69.00	67.50	50.00	49.00	69.00
TQA	8k	87.89	18.69	88.44	88.55	88.86	**89.63**	89.36	12.28	74.75	88.65
SMS	6k	41.70	32.46	43.76	**43.35**	42.21	44.25	43.02	13.45	40.38	41.59
PSC	11k	2.10	2.67	0.00	1.50	2.68	1.05	2.50	**4.57**	0.64	2.09
PSR	9k	**9.00**	3.77	4.50	4.50	4.62	3.79	8.50	3.50	2.50	6.50
LCC	1k	**68.22**	63.64	68.47	55.05	56.78	54.06	49.45	57.12	56.70	52.04
REP	4k	**61.73**	53.69	59.99	47.26	49.09	47.60	42.84	51.92	48.23	39.68
Avg.	7k	32.92	25.54	33.62	33.48	**35.32**	33.45	33.48	23.30	28.19	34.30

与基础模型相比，表现出显著的性能退化。相比之下，NTK、PI 和 YaRN 方法成功地在 32k 上维持了其性能，这些方法之间取得了可比的结果。这表明 PI 和 YaRN 在下游任务中表现相似，而 NTK 系列模型保持稳定。

值得注意的是，利用LoRA的LongLoRA方法相对于基础检查点LLaMA2也出现了性能下降。我们认为这可能是由于LongLoRA的训练过程较为敏感，我们在局限性讨论部分承认了这一点。

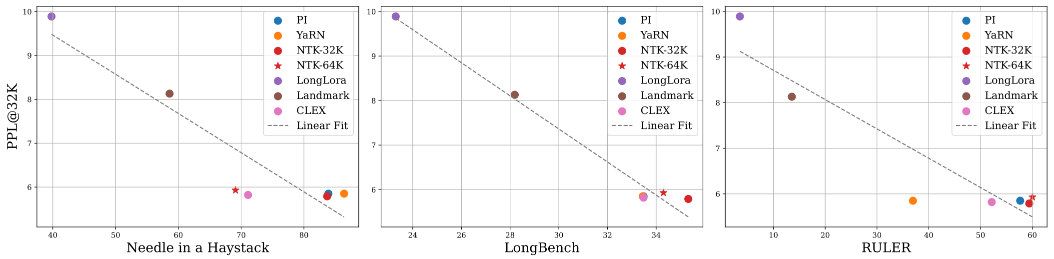

Figure 4: Perplexity and averaged downstream task accuracy for Needle in a haystack, LongBench and RULER.

Furthermore, the overall performance on LongBench has not shown significant improvement over LLaMA2.We hypothesize that this is due to the average length of LongBench test data (approximately 7.5k) being considerably shorter than the 32k context window of the long-context methods.

Many-shot In-Context Learning with Trec News We evaluate TREC News [Li and Roth, 2002] with 1 to 1000 in-context examples. Performance improves with more examples in Figure 2. Exact Attention methods show significant gains from 10 to 50 examples (+44.0%) and 100 to 1000 examples (+25.9%), with slower growth from 50 to 100 examples (+5.7%). Approximate Attention methods consistently underperform. Performance gains align with model perplexity in longer contexts; NTK-Frozen excels with fewer examples but underperforms with more.

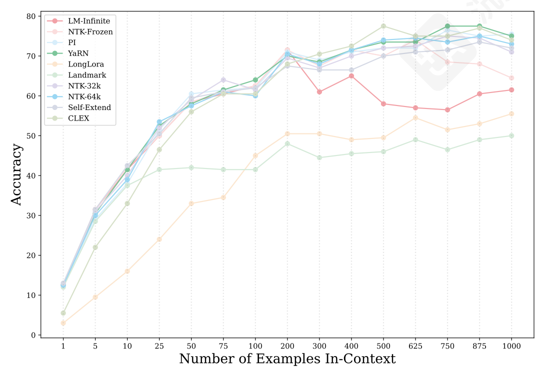

Figure 2: Many-shot in-context learning accuracy on TREC News.

6 Analysis

Perplexity and Downstream Tasks Several existing studies [Sun et al., 2021, An et al., 2023] suggest that perplexity may not consistently correlate with performance on long-range tasks. In Figure 4, we plot the perplexity of models from our evaluated benchmarks against their performance. The figure shows a general correlation between perplexity and model performance across various tasks for exact attention methods. However, approximate attention methods, LongLora, and Landmark on RULER, exhibit slight deviations from this trend but still roughly fit into the linear relationship. We hypothesize that previous studies mentioned this disconnection due to the lack of controlled studies and the presence of noisy data.

As depicted in Figure 1, although LM-Infinite achieves a good perplexity score at 32k, it fails to generalize beyond 4k. This limitation arises because the LM-Infinite method focuses on a context window of only 4k, resulting in poor retrieval ability for longer contexts. This suggests that we should consider downstream tasks at different lengths when evaluating perplexity for approximate attention methods.

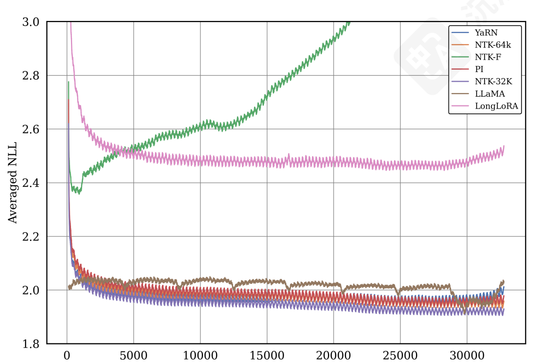

Figure 3: Averaged negative log-likelihood of different models broken down by context position.

Context extension hurts in the short term and gains in the long term While context extension seems to improve perplexity, in Table 4, we do not notice a significant gain in performance.

We hypothesize that while this dataset contains long tasks, the average length is much shorter than 32K. These methods seem to improve the ability to model language over the long term but hurt in the short term. To understand this better we

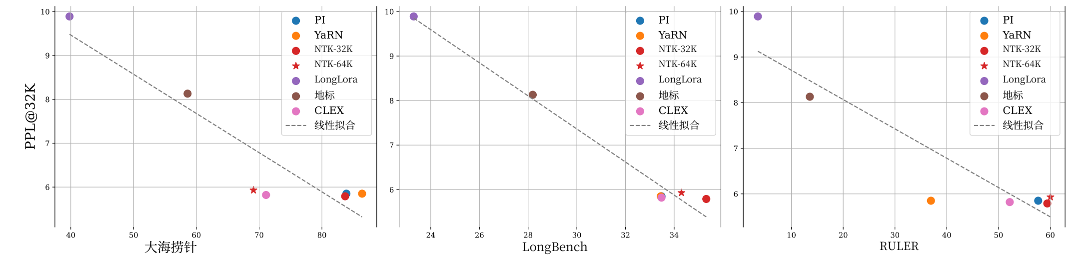

图4：大海捞针、LongBench 和 RULER 的困惑度及下游任务平均准确率。

此外，在LongBench上的整体性能并未显示出相对于LLaMA2的显著提升。我们推测这是由于LongBench测试数据的平均长度（约7.5k）远短于长上下文方法（32k）的上下文窗口。

多示例情境学习与TREC News 我们评估了TREC News [Li和Roth,2002] 使用1到1000个情境示例。在图2中，性能随着示例数量的增加而提高。精确注意力方法在10到50个示例（+44.0%）和100到1000个示例（+25.9%）之间表现出显著提升，但在50到100个示例（+5.7%）之间的增长较慢。近似注意力方法始终表现不佳。性能提升与较长上下文中的模型困惑度一致；NTK-Frozen在较少示例时表现优异，但在更多示例时表现不佳。

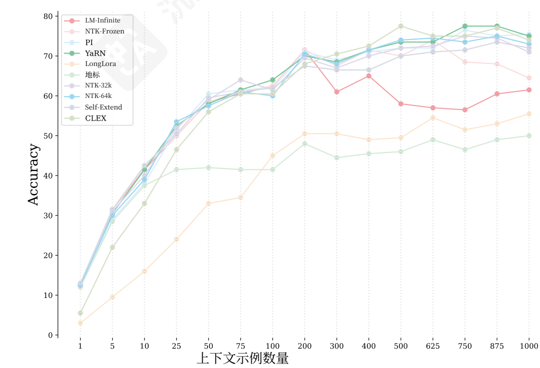

图2：TREC News 的多示例上下文学习准确率。

6 分析

Perplexit y 和下游任务 几个现有研究 [孙等人，2021年，安等人， 2023] 表明困惑度可能不会始终与长距离任务的性能相关。在图4中，我们绘制了评估基准中模型的困惑度与其性能的关系。该图显示了精确注意力方法在各种任务中困惑度与模型性能之间的普遍相关性。然而，近似注意力方法、LongLora和RULER上的Landmark表现出与这一趋势的轻微偏差，但仍大致符合线性关系。我们假设先前研究中提到的这种脱节是由于缺乏控制研究和存在噪声数据所致。

如图1所示，尽管LM-Infinite在32k时取得了良好的困惑度得分，但它无法泛化到4k以上。这种局限性是因为LM-Infinite方法仅关注4k的上下文窗口，导致对较长上下文的检索能力较差。这表明我们在评估近似注意力方法的困惑度时，应考虑不同长度的下游任务。

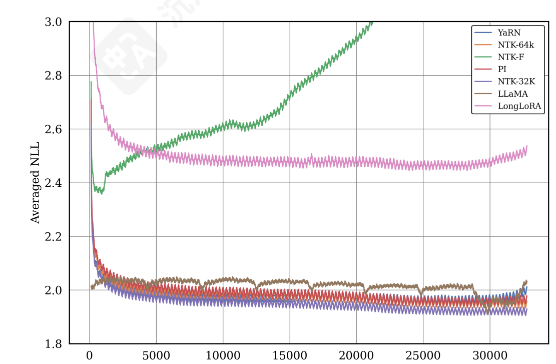

图3：按上下文位置分解的不同模型的平均负对数似然。

上下文扩展短期内有害，长期内有益虽然上下文扩展似乎提高了困惑度，但在表4中，我们没有注意到性能有显著提升。

我们假设，虽然这个数据集包含长任务，但平均长度远短于32K。这些方法似乎要 prove the 具备长期建模语言的能力，但在短期内会受损。为了更好地理解这一点，我们

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compute the averaged negative likelihood of each position of YaRN, LLaMa2, and NTK-32K per position (with LLaMa2 seeing just tokens every 4k chunks) in Figure 3.

NTK Generalizes Beyond 32k In Figure 1, we observe that NTK-32K successfully generalizes to unseen sequence lengths beyond 32k in both Needle-in-the-Haystack and RULER tasks, performing on par with NTK-64K. In contrast, NTK-F demonstrates generalization up to 8k but fails to extend further. This suggests that while NTK methods may possess the capability to generalize to longer unseen sequences, their effectiveness is contingent upon conditions such as continual fine-tuning.

We find that up until 4K they all improve as expected with LLaMa2 having the best NLL. After 4K they all fluctuate in average, but we see a clear separation with Yarn and NTK taking into account the long context. At extremely long context NTK remains a strong model whereas Yarn becomes reverts to a similar performance as LLaMa2.

7 Limitations and Broader Impacts

Limitations As we are limited by computing budget, we only experiment with Llama-2-7B as our base model. The findings in this paper may not generalize to other, possibly larger, base models. We only fine-tuned models to context sizes of 32k, and generalization behaviors to longer contexts may differ when training contexts are longer. We also acknowledge that the standardized training recipe with fixed hyperparameters may bias some models more than the other models.

Broader Impacts This paper provides an in-depth evaluation of various methods for extending the context lengths of LLMs, offering insights into their performance in long-range settings. By standardizing evaluations with a consistent base model and extension data, this work provides clearer benchmarks for future work. We also open-source code and models to promote transparency. Our models share similar risks as standard LLMs, where they may be used to generate harmful content and misinformation.

8 Conclusion

In this paper, we used a standardized approach to assess the performance of various long-context methods in LLMs. Our study underscores the role of perplexity as a crucial, performance indicator at length and highlights the trade-offs inherent in different attention mechanisms. We shed light on the strengths and weaknesses of various approaches, providing valuable insights for future research. As part of our commitment to open science, all our resources, including codebases, models, and checkpoints, will be open-sourced upon acceptance, fostering future advancements in this pivotal area of AI research.

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计算YaRN、LLaMA2和NTK-32K每个位置的加权负似然（其中LLaMA2仅看到每4k块中的标记）如图3所示。

<样式 id='1'>NTK 泛化超越 32k </样式>在图1中，我们观察到NTK-32K在寻找针和RULER任务中均成功泛化到32k以上的未见序列长度，表现与NTK-64K相当。相比之下，NTK-F泛化到8k但无法进一步扩展。这表明虽然NTK方法可能具备泛化到更长未见序列的能力，但其有效性取决于持续微调等条件。

我们发现，直到4k，它们都按预期提升，其中Llama2的NLL表现最佳。超过4k后，它们平均值波动，但考虑到长上下文，YaRN和NTK表现出明显差异。在极长上下文中，NTK仍为强模型，而YaRN性能则回归至与LLaMA2相似的水平。

7 局限性与更广泛的影响

局限性 由于受计算预算限制，我们仅以 Llama-2-7B 作为基础模型进行实验。本文的发现可能无法泛化到其他可能更大的基础模型。我们仅将模型微调到 32k 的上下文大小，当训练上下文更长时，泛化行为可能会有所不同。我们还承认，使用固定超参数的标准训练配方可能会使某些模型比其他模型更容易产生偏差。

更广泛的影响 本文深入评估了扩展大型语言模型上下文长度的各种方法，并对其在长距离环境中的性能提供了见解。通过使用一致的基座模型和扩展数据来标准化评估，这项工作为未来的工作提供了更清晰的基准。我们还开源代码和模型以促进透明度。我们的模型与标准大型语言模型具有相似的风险，可能被用于生成有害内容和错误信息。

8 结论

在本文中，我们使用标准化的方法评估了大型语言模型中各种长上下文方法的性能。我们的研究强调了困惑度作为长度关键性能指标的作用，并突出了不同注意力机制中固有的权衡。我们阐明了各种方法的优缺点，为未来的研究提供了宝贵的见解。作为我们对开放科学承诺的一部分，所有我们的资源，包括代码库、模型和检查点，将在接受后开源，促进这一关键人工智能研究领域的未来进步。

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Appendix

9.1 Phi-2 for Context Extension

We use another open-weight model, Phi-2-base Javaheripi et al. [2023] as the base point for context extension, to verify whether the trends and analyses we observed are consistent across different base models. Using an identical training recipe, we re-train and re-evaluate seven models with Phi-2-base.

9.1.1 Perplexity on Proof-file of Phi-2-base

We evaluate the perplexity score of Phi-2-base on seven models in Table 5. Consistent with our observations in the original submission, continuous fine-tuning methods like PI, YaRN, and LongLora effectively maintain low perplexity scores within the pre-training context length. However, perplexity scores escalate once the context length exceeds the pre-trained window. Notably, only NTK could generalize to unseen sequence lengths during both pre-training and continual fine-tuning.

Table 5: Perplexity results of different methods on Proof-file with Phi-2-base. NTK-32K and NTK-64K refer to NTK-Dynamic, which requires finetuning on longer text. Len refers to the longest-length examples seen at training or fine-tuning. Ex refers to the exact attention. All results are produced by our experiments.

	Model Details			Eval Length					
	Len	Ex	Methods	2k	4k	8k	16k	32k	64k
Frozen	2k	✓	Phi-2-base	**4.02**	25.72	175.05	-	-	-
	2k	✓	NTK-Frozen	**4.02**	3.73	4.07	5.49	12.58	36.68
	2k		Self-Extend	4.08	**3.70**	**3.48**	3.42	3.48	3.73
Finetuned	32k	✓	PI	7.53	6.75	6.25	5.97	5.83	45.00
	32k	✓	NTK-32K	4.24	3.81	3.51	**3.32**	**3.18**	**3.20**
	32k	✓	CLEX	5.53	4.32	3.78	3.51	3.42	3.60
	64k	✓	NTK-64K	4.63	4.14	3.82	3.61	3.47	3.38

9.1.2 RULER of Phi-2-base

We test all models on 12 diverse tasks (except QA-2) from the four Ruler Hsieh et al. [2024] categories in Table 6. Consistently, NTK-32k maintains relatively strong performance compared to other methods fine-tuned with a length cap of 32k. The only exception was CLEX and NTK-32k at 64k, showing a slight drop in performance.

Table 6: RULER evaluation on seven methods with Phi-2-base. Performance of models evaluated at length from 2k to 64k. Each score is computed by averaging the accuracy of 12 tasks. Train Len refers to the longest-length examples seen at continuous finetuning.

	Models	**Train Len**	**2k**	**4k**	**8k**	**16k**	**32k**	**64k**
Frozen	Phi-2-base	2k	83.73	-	-	-	-	-
	NTK-Frozen	2k	**83.98**	52.95	18.09	4.07	0.06	0.00
	Self-Extend	2k	68.55	50.82	36.65	22.00	7.83	2.32
Finetuned	PI	32k	25.51	23.19	16.88	14.99	4.78	0.00
	NTK-32K	32k	81.18	66.90	52.57	**46.53**	**32.06**	12.84
	CLEX	32k	75.33	**72.66**	**53.56**	46.23	25.46	13.03
	NTK-64K	64k	78.73	59.87	47.56	41.87	25.66	**17.69**

附录

9.1 Phi-2forContext 扩展

我们使用另一个开放权重模型 Phi-2-base Javaheripi 等人 [2023] 作为上下文扩展的基点，以验证我们观察到的趋势和分析是否在不同基础模型上一致。使用相同的训练配方，我们重新训练和重新评估了七个 Phi-2-base 模型。

9.1.1 Phi-2-base 证明文件的困惑度

我们在表 5 中的七个模型上评估了 Phi-2-base 的困惑度得分。与我们在原始提交中的观察一致，持续微调方法如 PI、YaRN 和 LongLora 在预训练上下文长度内有效维持了低困惑度得分。然而，一旦上下文长度超过预训练窗口，困惑度得分就会上升。值得注意的是，只有 NTK 在预训练和持续微调期间都能泛化到未见序列长度。

表 5：Phi-2-base 在证明文件上不同方法的困惑度结果。NTK-32K 和 NTK-64K 指的是 NTK-Dynamic，它需要在更长的文本上进行微调。Len 指的是训练或微调时见到的最长长度的示例。Ex 指的是精确注意力。所有结果都是由我们的实验产生的。

	模型详情			评估长度					
	Len	Ex	方法	2k	4k	8k	16k	32k	64k
冻结	2k	✓	Phi-2-base	**4.02**	25.72	175.05	-	-	-
	2k	✓	NTK-Frozen	**4.02**	3.73	4.07	5.49	12.58	36.68
	2k		Self-Extend	4.08	**3.70**	**3.48**	3.42	3.48	3.73
微调	32k	✓	PI	7.53	6.75	6.25	5.97	5.83	45.00
	32k	✓	NTK-32K	4.24	3.81	3.51	**3.32**	**3.18**	**3.20**
	32k	✓	CLEX	5.53	4.32	3.78	3.51	3.42	3.60
	64k	✓	NTK-64K	4.63	4.14	3.82	3.61	3.47	3.38

9.1.2 Phi-2-base的RULER

我们在表6中列出的四种Ruler Hsieh 等人 [2024] 类别中的12项不同任务（除QA-2外）上测试了所有模型。始终如一地，NTK-32k与其他使用32k长度限制进行fine-tuned的方法相比，保持了相对较强的性能。唯一的例外是CLEX和NTK-32k在64k时，性能略有下降。

表6：使用Phi-2-base对七种方法进行RULER评估。模型在长度从2k到64k时的性能。每个分数通过平均12个任务的准确率计算得出。TrainLen指的是在连续微调中遇到的最长长度的示例。

	模型	**训练长度**	**2k**	**4k**	**8k**	**16k**	**32k**	**64k**
冻结	Phi-2-base	2k	83.73	-	-	-	-	-
	NTK-Frozen	2k	**83.98**	52.95	18.09	4.07	0.06	0.00
	Self-Extend	2k	68.55	50.82	36.65	22.00	7.83	2.32
微调	PI	32k	25.51	23.19	16.88	14.99	4.78	0.00
	NTK-32K	32k	81.18	66.90	52.57	**46.53**	**32.06**	12.84
	CLEX	32k	75.33	**72.66**	**53.56**	46.23	25.46	13.03
	NTK-64K	64k	78.73	59.87	47.56	41.87	25.66	**17.69**

9.2 Implementation Details

Training To maintain consistency across all models, we use a fixed training protocol [Chen et al., 2023b]. We adopt standard practices by applying an exponential moving average (EMA) to the model weights with a constant decay rate. The majority of our training hyperparameters are derived from [Chen et al., 2023b], including a learning rate of 2×10^{-5}. We implement a linear warm-up for the learning rate and set the weight decay to zero, utilizing 8 NVIDIA A100 GPUs.

For LongLora, we fine-tune the LoRA adapter weights along with trainable embeddings and normalization, subsequently integrating these trained weights into the LLaMA2 base model for evaluation. For Landmark Attention, the training context length is 512, with a block size of 64. For CLEX, we set the max scale factor to 32 and use the SiLU activation function. We present the hyperparameter settings for different methods during the training stage in Table 7.

Inference We list the scale factors used for different length ranges during inference in Table 8. For NTK-RoPE, given the maximum observed length during training or inference, C_{test}, and the scaling hyperparameter s, we follow Fu et al. [2024] in replacing t with $s \cdot \frac{\max(C', C_{\text{test}})}{C} - (s-1)$, and set the hyperparameter s to $\frac{C'}{2C}$ during both training and inference. For LM-infinite, we set the global memory $G = 10$ and the local window $M = 4096$. For Landmark Attention, the training context length is set to 512, with a block size of 64. For Self-Extend, we set the local window size M for neighbor tokens to 1024 and the group size N to 64.

Table 7: Hyperparameters for Different Long Sequence Methods in Training.

Models	Train Len	Train Tokens	α	bsz	lr
PI	32k	1B	8.0	32	2e-5
NTK-32K	32k	1B	29.0	32	2e-5
YaRN	32k	1B	8.0	32	2e-5
LongLora	32k	1B	8.0	32	2e-5
Landmark	32k	1B	-	32	2e-5
NTK-64K	64k	1B	57.0	32	2e-5

Table 8: Hyperparameters for the Scale Factor Different Long-context Methods in Inference.

Models	4k	8k	16k	32k	64k
Llama2	-	-	-	-	-
LM-Infinite	-	-	-	-	-
NTK-Frozen	1.0	3.0	7.0	15.0	31.0
PI	8.0	8.0	8.0	8.0	8.0
NTK-32K	29.0	29.0	29.0	29.0	61.0
YaRN	8.0	8.0	8.0	8.0	8.0
LongLora	8.0	8.0	8.0	8.0	8.0
Landmark	-	-	-	-	-
NTK-64K	57.0	57.0	57.0	57.0	57.0

9.3 Training Data Construction

We sample 1B tokens from a long-context data mixture following Fu et al. [2024]. We use the SlimPajama [Soboleva et al., 2023] dataset for continuous finetuning. This dataset serves as an open-source replication of the LLaMA [Touvron et al., 2023] pretraining data mixture. It comprises 82% web data (sourced 67% from CommonCrawl and 15% from C4), 4.5% code data (Github), 4.5% Wikipedia content, 4.5% books, 2.5% Arxiv papers, and 2.0% StackExchange content. We use per-source length-upsampling to sample 1B tokens from the datasets, which increases the portion of long sequences while keeping the domain mixture the same. We packed all sampled data into

9.2 实现细节

训练 为了在所有模型中保持一致性，我们使用固定的训练协议 [陈等人，2023b]。我们采用标准做法，通过应用具有恒定衰减率的指数移动平均（EMA）到模型权重。我们的大多数训练超参数来自[陈等人，2023b]，，包括学习率为 2×10^{-5}。我们对学习率实施线性预热，并将权重衰减设置为零，利用8个NVIDIA A100 GPU。

对于LongLora，我们微调LoRA适配器权重以及可训练嵌入和归一化，随后将这些训练好的权重集成到LLaMA2基础模型中进行评估。对于Landmark注意力，训练上下文长度为512，块大小为64。对于CLEX，我们将最大缩放因子设置为32，并使用SiLU激活函数。我们在表7中展示了不同方法在训练阶段的超参数设置。

推理我们在表8中列出了用于不同长度范围推理时的缩放因子。对于NTK-RoPE，给定训练或推理过程中观察到的最大长度 C_{test}以及缩放超参数s，我们遵循Fu等人 [2024] 的方法，将 t 替换为 $s \cdot \frac{\max(C', C_{\text{test}})}{C} - (s-1)$，并在训练和推理过程中将超参数$s$ 设置为 $\frac{C'}{2C}$。对于LM-infinite，我们设置了全局内存 $G = 10$ 和本地窗口 $M = 4096$。对于Landmark注意力，训练上下文长度设置为512，块大小为64。对于Self-Extend，我们将邻居token的本地窗口大小 M 设置为1024，并将组大小 N 设置为64。

表7：训练中不同长序列方法的超参数。

模型	训练长度	训练 Token数	α	bsz	lr
PI	32k	1B	8.0	32	2e-5
NTK-32K	32k	1B	29.0	32	2e-5
YaRN	32k	1B	8.0	32	2e-5
LongLora	32k	1B	8.0	32	2e-5
地标	32k	1B	-	32	2e-5
NTK-64K	64k	1B	57.0	32	2e-5

表8：推理中不同长上下文方法的缩放因子超参数。

模型	4k	8k	16k	32k	64k
Llama2	-	-	-	-	-
LM-Infinite	-	-	-	-	-
NTK-Frozen	1.0	3.0	7.0	15.0	31.0
PI	8.0	8.0	8.0	8.0	8.0
NTK-32K	29.0	29.0	29.0	29.0	61.0
YaRN	8.0	8.0	8.0	8.0	8.0
LongLora	8.0	8.0	8.0	8.0	8.0
Landmark	-	-	-	-	-
NTK-64K	57.0	57.0	57.0	57.0	57.0

9.3 训练数据构建

我们从 Fu 等人 [2024] 的长上下文数据混合中采样 1B 个 token。我们使用 SlimPajama [Soboleva 等人， 2023] 数据集进行连续微调。该数据集是 LLaMA [Touvron 等人， 2023] 预训练数据混合的开源复制品。它包含 82% 的网络数据（其中 67% 来自 CommonCrawl，15% 来自 C4），4.5% 的代码数据（来自 Github），4.5% 的维基百科内容，4.5% 的书籍，2.5% 的 Arxiv 论文，以及 2.0% 的 StackExchange 内容。我们使用按源长度上采样方法从数据集中采样 1B 个 token，这增加了长序列的比例，同时保持了领域混合不变。我们将所有采样数据打包成

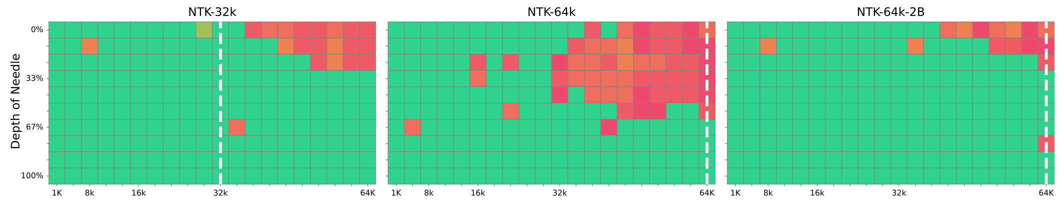

Figure 5: Needle in a Haystack evaluation. "NTK-64-2B" represents the NTK-64K model trained with 2B tokens. Green squares indicates a high retrieval success rate, the white dashed line denotes the longest length examples seen at training or finetuning, and the Y-axis represents the distance to the retrieved target.

chunks of the corresponding training length, regardless of document boundaries, following common practiceTouvron et al. [2023], Fu et al. [2024].

9.4 Longer Model Needs more Training Tokens

We observe that the performance of NTK-64K is not as good as NTK-32K. Consequently, we further sample 2B tokens from a long-context data mixture from Fu et al. [2024] for training and evaluate the model on the "Needle in A Haystack" task, as shown in Figure 5. Our NTK-64K model demonstrates a significant performance improvement when trained with more tokens, indicating that longer models require more tokens for effective training.

9.5 RoPE Scale Factor for Dynamic NTK

We observe that the scale factor significantly degrades NTK-Dynamic models, particularly causing performance deterioration in shorter sequences. Therefore, we conduct a grid search to determine a better scale factor for different input lengths. The scale factor and its relationship with perplexity on PG19 are reported in Table 9.

Table 9: The scale factor and its relationship with perplexity on PG19. We only use the first 2 documents of PG19 to calculate the perplexity.

Models	Scale Factor	4k	8k	16k	32k	64k
NTK-Frozen	1	**7.65**	118.82	NaN	NaN	NaN
	3	8.19	**7.99**	57.15	386.02	NaN
	7	9.39	9.26	**9.61**	72.62	486.13
	15	11.53	12.04	12.98	**20.15**	180.59
	31	16.18	20.66	26.67	40.06	**69.01**
	63	30.22	48.78	69.89	89.75	118.59
NTK-32K	1	12.64	NaN	NaN	NaN	NaN
	5	7.84	7.638	10.36	NaN	NaN
	13	**7.686**	7.459	7.25	8.35	NaN
	29	7.689	**7.457**	**7.24**	**6.82**	9.11
	61	7.8	7.565	7.34	6.91	**6.63**
	125	7.99	7.774	7.57	7.13	6.83
NTK-64K	1	19.16	NaN	NaN	NaN	NaN
	9	8.02	7.79	7.63	22.6	NaN
	25	**7.89**	**7.65**	**7.443**	7.04	14.02
	57	7.922	7.67	7.44	**7.01**	**6.75**
	121	8.016	7.75	7.51	7.06	6.77

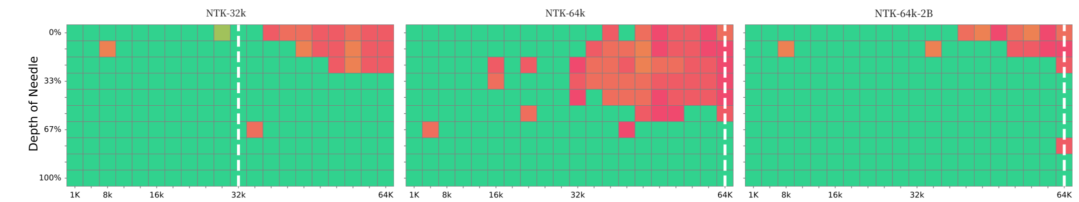

图5：大海捞针评估。“NTK-64-2B”表示使用2B个token训练的NTK-64K模型。绿色方块表示检索成功率较高，白色虚线表示训练或微调时遇到的最长长度示例，Y轴表示检索目标距离。

对应训练长度的块，无论文档边界如何，遵循常见做法 Touvron 等人 [2023], Fuet al. [2024]。

9.4 更长模型需要更多训练token

我们观察到 NTK-64K 的性能不如 NTK-32K。因此，我们从 Fu 等人 [1] 的长上下文数据混合中进一步采样2B个 token 用于训练，并在“大海捞针”任务上评估模型，如图 5 所示。我们的 NTK-64K 模型在用更多 token 训练时表现出显著的性能提升，表明更长的模型需要更多 token 才能有效训练。

9.5 RoPE 缩放因子用于动态 NTK

我们观察到缩放因子会显著降低NTK-Dynamic模型，特别是在较短的序列中导致性能下降。因此，我们进行了网格搜索，以确定不同输入长度下的更好缩放因子。缩放因子及其与PG19上的困惑度的关系在表9中报告。

表 9：缩放因子及其与 PG19 上困惑度的关系。我们仅使用 PG19 的前 2 篇文档来计算困惑度。

模型	缩放因子	4k	8k	16k	32k	64k
NTK-Frozen	1	**7.65**	118.82	NaN	NaN	NaN
	3	8.19	**7.99**	57.15	386.02	NaN
	7	9.39	9.26	**9.61**	72.62	486.13
	15	11.53	12.04	12.98	**20.15**	180.59
	31	16.18	20.66	26.67	40.06	**69.01**
	63	30.22	48.78	69.89	89.75	118.59
NTK-32K	1	12.64	NaN	NaN	NaN	NaN
	5	7.84	7.638	10.36	NaN	NaN
	13	**7.686**	7.459	7.25	8.35	NaN
	29	7.689	**7.457**	**7.24**	**6.82**	9.11
	61	7.8	7.565	7.34	6.91	**6.63**
	125	7.99	7.774	7.57	7.13	6.83
NTK-64K	1	19.16	NaN	NaN	NaN	NaN
	9	8.02	7.79	7.63	22.6	NaN
	25	**7.89**	**7.65**	**7.443**	7.04	14.02
	57	7.922	7.67	7.44	**7.01**	**6.75**
	121	8.016	7.75	7.51	7.06	6.77

9.6 LongLora Validation

To validate our LongLora Chen et al. [2023b] implementation, we reproduced their Llama-2-7b-longlora-32k model following LongLora's training data and training recipe. We evaluated the perplexity for the corresponding length on PG19 and Proof-file in Table 10.

Table 10: Perplexity results of LongLora reported and our reproduction on PG 19 and Proof-file.

Method	2k	4k	8k	16k	32k
PG19					
Llama-2-7b-longlora-32k	8.29	7.83	7.54	7.35	7.22
Our Reproduction	8.10	7.69	7.43	7.28	7.32
Proof-file					
Llama-2-7b-longlora-32k	3.35	3.01	2.78	2.61	2.50
Our Reproduction	3.33	3.01	2.80	2.67	2.61

9.7 LongBench Results

The evaluation results of all methods on LongBench are presented in Table 11.

Table 11: LongBench results. N-32 and N-64 refer to NTK finetuned on 32K and 64K context lengths respectively. Inf refers to LM-Infinite. SE refers to Self-Extend. LLR refers to LongLora. AvgLen refers to average length of the datasets. Train Len refers to the longest length examples seen at training or finetuning. Eval Len refers to the maximum length of the input prompt. ✓refers to whether the method is exact attention.

Exact	AvgLen	Frozen				Finetuned						
		Base ✓	Inf	N-F ✓	SE	PI ✓	N-32 ✓	YaRN ✓	CLEX	LLR	Land ✓	N-64
Train Len		4k	4k	4k	4K	32k	32k	32k	32k	32k	32k	64k
Eval Len		4k	32k	32k	32K	32k	32k	32k	32k	32k	32k	32k
NQA	**18,409**	21.09	10.39	3.88	23.49	23.02	23.73	19.82	24.19	12.07	12.47	**24.31**
QAPR	3,619	26.94	22.58	26.79	28.75	25.85	**27.50**	26.98	23.36	20.15	19.06	24.97
MFQA	4,559	32.42	26.19	29.82	32.66	35.10	38.22	37.11	40.83	24.50	21.86	**40.60**
HPQA	9,151	31.23	16.13	32.10	37.63	36.98	**41.56**	38.60	35.59	27.41	33.66	41.47
WMQA	4,887	25.75	20.64	22.34	30.70	29.38	**31.58**	30.63	28.24	21.46	24.94	28.62
MSQ	11,214	10.55	5.26	8.84	15.73	16.80	17.41	**22.08**	17.12	11.46	11.41	18.24
GR	8,734	17.32	13.43	17.87	13.15	25.61	**28.27**	20.98	24.68	24.05	17.20	24.37
QMSM	10,614	21.28	6.10	15.35	20.20	21.19	21.52	20.66	21.55	17.66	18.83	**21.65**
MNWS	2,113	3.44	3.63	9.30	1.50	10.55	22.13	8.91	16.96	21.19	19.43	**25.02**
TREC	5,177	66.00	61.00	67.50	69.00	**71.00**	69.00	69.00	67.50	50.00	49.00	69.00
TRVQA	8,209	87.89	81.40	18.69	88.44	88.55	88.86	**89.63**	89.36	12.28	74.75	88.65
SMSM	6,258	41.70	15.07	32.46	43.76	**43.35**	42.21	44.25	43.02	13.45	40.38	41.59
PSC	11,141	2.10	1.62	2.67	0.00	1.50	2.68	1.05	2.50	**4.57**	0.64	2.09
PSR	9,289	**9.00**	4.00	3.77	4.50	4.50	4.62	3.79	8.50	3.50	2.50	6.50
LCC	1,235	**68.22**	67.68	63.64	68.47	55.05	56.78	54.06	49.45	57.12	56.70	52.04
REPO	4,206	**61.73**	58.27	53.69	59.99	47.26	49.09	47.60	42.84	51.92	48.23	39.68
Average	7,425	32.92	25.84	25.54	33.62	33.48	**35.32**	33.45	33.48	23.30	28.19	34.30

9.8 RULER Subtasks Result

The performance of models on different lengths and breakdowns by 13 subtasks are reported in Table 12(RULER on 4k), Table 13(RULER on 8k), Table 14(RULER on 16k), Table 15(RULER on 32k) and Table 16(RULER on 64k).

9.6 LongLora 验证

为了验证我们的LongLora Chen等人 [2023b] 实现，我们根据LongLora的训练数据和训练配方复现了他们的Llama-2-7b-longlora-32k模型。我们在表10中评估了PG19和证明文件上对应长度的困惑度。

表 10：LongLora 报告的困惑度结果和我们在 PG 19 和证明文件上的复制品。

方法	2k	4k	8k	16k	32k
PG19					
Llama-2-7b-longlora-32k	8.29	7.83	7.54	7.35	7.22
我们的复制品	8.10	7.69	7.43	7.28	7.32
证明文件					
Llama-2-7b-longlora-32k	3.35	3.01	2.78	2.61	2.50
我们的复制品	3.33	3.01	2.80	2.67	2.61

9.7 LongBench 结果

所有方法在 LongBench 上的评估结果在表 11 中呈现。

表 11：LongBench 结果。N-32 和 N-64 分别指在 32k 和 64k 上下文长度上微调的 NTK。Inf 指 LM-Infinite。SE 指 Self-Extend。LLR 指 LongLora。AvgLen 指数据集的平均长度。Train Len 指训练或微调时遇到的最长长度示例。Eval Len 指输入提示的最大长度。 ✓ 指该方法是否为精确注意力。

精确	平均长度	冻结				微调						
		Base ✓	Inf	N-F ✓	SE	PI ✓	N-32 ✓	YaRN ✓	CLEX	LLR	Land ✓	N-64
训练长度		4k	4k	4k	4K	32k	32k	32k	32k	32k	32k	64k
评估长度		4k	32k	32k	32K	32k	32k	32k	32k	32k	32k	32k
NQA	**18,409**	21.09	10.39	3.88	23.49	23.02	23.73	19.82	24.19	12.07	12.47	**24.31**
QAPR	3,619	26.94	22.58	26.79	28.75	25.85	**27.50**	26.98	23.36	20.15	19.06	24.97
MFQA	4,559	32.42	26.19	29.82	32.66	35.10	38.22	37.11	40.83	24.50	21.86	**40.60**
HPQA	9,151	31.23	16.13	32.10	37.63	36.98	**41.56**	38.60	35.59	27.41	33.66	41.47
WMQA	4,887	25.75	20.64	22.34	30.70	29.38	**31.58**	30.63	28.24	21.46	24.94	28.62
MSQ	11,214	10.55	5.26	8.84	15.73	16.80	17.41	**22.08**	17.12	11.46	11.41	18.24
GR	8,734	17.32	13.43	17.87	13.15	25.61	**28.27**	20.98	24.68	24.05	17.20	24.37
QMSM	10,614	21.28	6.10	15.35	20.20	21.19	21.52	20.66	21.55	17.66	18.83	**21.65**
MNWS	2113	3.44	3.63	9.30	1.50	10.55	22.13	8.91	16.96	21.19	19.43	**25.02**
TREC	5,177	66.00	61.00	67.50	69.00	**71.00**	69.00	69.00	67.50	50.00	49.00	69.00
TRVQA	8,209	87.89	81.40	18.69	88.44	88.55	88.86	**89.63**	89.36	12.28	74.75	88.65
SMSM	6,258	41.70	15.07	32.46	43.76	**43.35**	42.21	44.25	43.02	13.45	40.38	41.59
PSC	11,141	2.10	1.62	2.67	0.00	1.50	2.68	1.05	2.50	**4.57**	0.64	2.09
PSR	9,289	**9.00**	4.00	3.77	4.50	4.50	4.62	3.79	8.50	3.50	2.50	6.50
LCC	1,235	**68.22**	67.68	63.64	68.47	55.05	56.78	54.06	49.45	57.12	56.70	52.04
REPO	4,206	**61.73**	58.27	53.69	59.99	47.26	49.09	47.60	42.84	51.92	48.23	39.68
平均值	7425	32.92	25.84	25.54	33.62	33.48	**35.32**	33.45	33.48	23.30	28.19	34.30

9.8 RULER 子任务结果

不同长度和按13个子任务分解的模型性能在表12(RULER on 4k)、表13(RULER on 8k)、表14(RULER on 16k)、表15(RULER on 32k)和表16(RULER on 64k)中报告。

Table 12: Ruler results on 4k context length. N-32 and N-64 refer to NTK finetuned on 32K and 64K context lengths respectively. Inf refers to LM-Infinite. SE refers to Self-Extend. LLR refers to LongLora. Train Len refers to the longest length examples seen at training or finetuning. Eval Len refers to the maximum length of the input prompt. ✓refers to whether the method is exact attention.

Exact	Frozen				Finetuned						
	Base	Inf	N-F	SE	PI	N-32	YaRN	CLEX	LLR	Land	N-64
	✓		✓		✓	✓	✓	✓			✓
Train Len	4k	4k	4k	4k	32k	32k	32k	32k	32k	32k	64k
Eval Len	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k
NIAH_S1	100.00	100.00	100.00	100.00	98.00	100.00	99.60	100.00	0.00	49.00	100.00
NIAH_S2	100.00	100.00	100.00	100.00	99.80	100.00	88.60	100.00	0.00	20.60	100.00
NIAH_S3	99.20	95.80	98.80	89.80	99.80	94.20	53.00	89.60	0.00	10.00	97.20
NIAH_M1	99.20	98.80	99.20	79.00	99.20	99.20	62.60	95.80	0.00	10.60	98.00
NIAH_M2	88.00	88.00	88.20	26.00	95.40	97.40	14.00	83.20	0.00	6.80	97.00
NIAH_M3	61.40	62.00	61.60	14.40	78.00	68.20	8.20	53.80	0.00	1.20	84.80
NIAH_MV	83.55	90.45	86.60	82.10	95.45	96.40	50.25	95.10	0.05	10.80	96.15
NIAH_MQ	95.45	96.15	96.00	90.70	96.95	97.00	62.00	96.20	0.00	5.35	98.25
VT	57.72	58.56	56.48	8.92	96.64	98.16	25.68	85.72	0.00	2.92	97.00
CWE	78.20	75.90	78.20	73.56	81.38	80.86	58.78	82.60	64.70	23.16	74.26
FWE	84.33	84.20	84.93	80.07	58.40	85.53	26.20	52.60	18.53	84.93	81.40
QA_1	62.20	60.40	62.40	60.60	57.80	62.40	60.20	55.80	26.20	37.20	55.80
QA_2	43.00	43.40	42.40	40.20	42.40	46.20	43.20	38.20	28.00	28.20	46.00
Avg.	80.94	81.05	81.14	65.03	84.56	86.58	50.18	79.12	10.58	22.37	86.60

Table 13: Ruler results on 8k context length.

Exact	Frozen				Finetuned						
	Base	Inf	N-F	SE	PI	N-32	YaRN	CLEX	LLR	Land	N-64
	✓		✓		✓	✓	✓	✓			✓
Train Len	4k	4k	4k	4k	32k	32k	32k	32k	32k	32k	64k
Eval Len	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k
NIAH_S1	-	46.00	61.60	100.00	99.00	99.80	100.00	100.00	0.00	46.00	100.00
NIAH_S2	-	36.60	59.40	98.80	100.00	100.00	99.40	100.00	0.00	7.20	100.00
NIAH_S3	-	20.80	51.00	88.60	99.20	94.20	96.00	97.80	0.00	3.80	99.20
NIAH_M1	-	27.80	46.00	69.40	98.00	94.20	86.60	90.20	0.00	7.60	95.20
NIAH_M2	-	4.40	11.00	8.20	91.60	86.20	60.60	66.00	0.00	1.60	86.60
NIAH_M3	-	2.60	4.00	3.20	48.40	52.20	34.60	11.80	0.00	0.00	47.40
NIAH_MV	-	30.35	41.35	52.95	65.50	85.95	70.40	61.25	0.00	6.25	84.75
NIAH_MQ	-	30.15	50.40	78.70	93.25	95.20	92.45	86.95	0.00	3.35	94.95
VT	-	4.88	69.88	1.48	91.20	96.16	77.52	48.16	0.00	3.08	94.36
CWE	-	65.08	40.30	30.82	45.66	45.76	44.72	32.72	18.92	22.08	40.80
FWE	-	56.73	64.87	59.00	65.07	70.13	10.53	45.40	16.73	76.60	54.13
QA_1	-	35.80	44.40	31.00	50.80	49.20	43.20	48.20	22.80	25.00	50.20
QA_2	-	29.00	33.60	37.40	40.80	41.80	36.80	42.60	24.40	25.20	44.80
Avg.	-	30.01	44.45	50.73	76.04	77.75	65.60	63.93	6.37	17.52	76.34

表12：在64k上下文长度上的RULER结果。N-32和N-64分别指在32K和64K上下文长度上微调的NTK。Inf指LM-Infinite。SE指Self-Extend。LLR指LongLora。Train Len指训练或微调时遇到的最长长度示例。Eval Len指输入提示的最大长度。✓指该方法是否为精确注意力。

精确	冻结				微调						
	Base	Inf	N-F	SE	PI	N-32	YaRN	CLEX	LLR	Land	N-64
	✓		✓		✓	✓	✓	✓			✓
Train Len	4k	4k	4k	4k	32k	32k	32k	32k	32k	32k	64k
Eval Len	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k
NIAH_S1	100.00	100.00	100.00	100.00	98.00	100.00	99.60	100.00	0.00	49.00	100.00
NIAH_S2	100.00	100.00	100.00	100.00	99.80	100.00	88.60	100.00	0.00	20.60	100.00
NIAH_S3	99.20	95.80	98.80	89.80	99.80	94.20	53.00	89.60	0.00	10.00	97.20
NIAH M1_	99.20	98.80	99.20	79.00	99.20	99.20	62.60	95.80	0.00	10.60	98.00
NIAH M2_	88.00	88.00	88.20	26.00	95.40	97.40	14.00	83.20	0.00	6.80	97.00
NIAH M3_	61.40	62.00	61.60	14.40	78.00	68.20	8.20	53.80	0.00	1.20	84.80
NIAH MV_	83.55	90.45	86.60	82.10	95.45	96.40	50.25	95.10	0.05	10.80	96.15
NIAH_MQ	95.45	96.15	96.00	90.70	96.95	97.00	62.00	96.20	0.00	5.35	98.25
VT	57.72	58.56	56.48	8.92	96.64	98.16	25.68	85.72	0.00	2.92	97.00
CWE	78.20	75.90	78.20	73.56	81.38	80.86	58.78	82.60	64.70	23.16	74.26
FWE	84.33	84.20	84.93	80.07	58.40	85.53	26.20	52.60	18.53	84.93	81.40
QA_1	62.20	60.40	62.40	60.60	57.80	62.40	60.20	55.80	26.20	37.20	55.80
QA_2	43.00	43.40	42.40	40.20	42.40	46.20	43.20	38.20	28.00	28.20	46.00
Avg.	80.94	81.05	81.14	65.03	84.56	86.58	50.18	79.12	10.58	22.37	86.60

表13：8k上下文长度上的RULER结果。

精确	冻结				微调						
	Base	Inf	N-F	SE	PI	N-32	YaRN	CLEX	LLR	Land	N-64
	✓		✓		✓	✓	✓	✓			✓
训练长度	4k	4k	4k	4k	32k	32k	32k	32k	32k	32k	64k
评估长度	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k
NIAH_S1	-	46.00	61.60	100.00	99.00	99.80	100.00	100.00	0.00	46.00	100.00
NIAH_S2	-	36.60	59.40	98.80	100.00	100.00	99.40	100.00	0.00	7.20	100.00
NIAH_S3	-	20.80	51.00	88.60	99.20	94.20	96.00	97.80	0.00	3.80	99.20
NIAH M1_	-	27.80	46.00	69.40	98.00	94.20	86.60	90.20	0.00	7.60	95.20
NIAH M2_	-	4.40	11.00	8.20	91.60	86.20	60.60	66.00	0.00	1.60	86.60
NIAH M3_	-	2.60	4.00	3.20	48.40	52.20	34.60	11.80	0.00	0.00	47.40
NIAH MV_	-	30.35	41.35	52.95	65.50	85.95	70.40	61.25	0.00	6.25	84.75
NIAH_MQ	-	30.15	50.40	78.70	93.25	95.20	92.45	86.95	0.00	3.35	94.95
VT	-	4.88	69.88	1.48	91.20	96.16	77.52	48.16	0.00	3.08	94.36
CWE	-	65.08	40.30	30.82	45.66	45.76	44.72	32.72	18.92	22.08	40.80
FWE	-	56.73	64.87	59.00	65.07	70.13	10.53	45.40	16.73	76.60	54.13
QA_1	-	35.80	44.40	31.00	50.80	49.20	43.20	48.20	22.80	25.00	50.20
QA_2	-	29.00	33.60	37.40	40.80	41.80	36.80	42.60	24.40	25.20	44.80
Avg.	-	30.01	44.45	50.73	76.04	77.75	65.60	63.93	6.37	17.52	76.34

Table 14: Ruler results on 16k context length.

Exact	Frozen				Finetuned						
	Base	Inf	N-F	SE	PI	N-32	YaRN	CLEX	LLR	Land	N-64
	✓		✓		✓	✓	✓	✓			✓
Train Len	4k	4k	4k	4k	32k	32k	32k	32k	32k	32k	64k
Eval Len	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k
NIAH_S1	-	21.00	14.20	99.80	97.20	99.40	100.00	99.80	0.00	42.40	99.80
NIAH_S2	-	17.00	17.40	93.40	100.00	100.00	99.20	100.00	0.20	6.80	100.00
NIAH_S3	-	11.60	8.20	77.00	99.60	98.60	89.60	99.60	0.00	3.60	100.00
NIAH_M1	-	15.80	9.20	60.00	97.80	93.20	83.40	89.40	0.00	5.60	90.80
NIAH_M2	-	0.00	0.60	3.80	82.80	79.80	19.60	72.00	0.00	0.80	67.60
NIAH_M3	-	1.00	0.00	1.80	34.20	18.20	7.40	15.00	0.00	0.00	29.60
NIAH_MV	-	8.40	6.90	38.85	77.55	81.95	58.75	62.40	0.00	4.80	83.50
NIAH_MQ	-	8.85	7.95	59.30	90.95	86.20	85.15	81.60	0.00	2.75	90.35
VT	-	6.56	11.28	1.16	68.84	83.56	47.12	48.16	0.00	2.52	88.68
CWE	-	19.94	28.36	17.80	27.26	26.32	23.72	28.60	0.62	11.90	21.20
FWE	-	77.13	25.80	59.80	47.93	61.73	10.13	57.33	12.93	81.60	51.73
QA_1	-	22.80	36.40	28.00	46.00	45.20	43.20	49.20	13.20	23.00	45.00
QA_2	-	24.20	26.00	31.60	35.20	36.00	37.40	33.40	20.80	26.20	36.00
Avg.	-	18.02	14.79	44.02	69.64	70.01	54.21	64.35	3.67	16.31	69.56

Table 15: Ruler results on 32k context length.

Exact	Frozen				Finetuned						
	Base	Inf	N-F	SE	PI	N-32	YaRN	CLEX	LLR	Land	N-64
	✓		✓		✓	✓	✓	✓			✓
Train Len	4k	4k	4k	4k	32k	32k	32k	32k	32k	32k	64k
Eval Len	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k
NIAH_S1	-	7.80	0.00	83.00	97.20	99.00	85.80	85.60	0.00	33.80	100.00
NIAH_S2	-	7.00	0.00	68.40	99.00	100.00	81.00	94.00	0.00	3.20	99.20
NIAH_S3	-	6.40	0.00	42.80	97.00	99.40	62.40	97.20	0.00	2.60	96.40
NIAH_M1	-	8.80	0.00	29.40	93.40	90.80	63.20	78.40	0.00	5.40	82.60
NIAH_M2	-	0.00	0.00	2.40	48.80	39.40	6.40	40.40	0.00	0.20	36.60
NIAH_M3	-	0.00	0.00	1.40	5.80	8.60	1.20	8.00	0.00	0.00	7.20
NIAH_MV	-	3.75	0.00	24.65	61.30	68.20	37.95	60.75	0.00	2.80	82.20
NIAH_MQ	-	2.05	0.00	20.35	68.65	78.25	46.95	67.80	0.05	2.35	85.80
VT	-	2.08	0.00	2.32	56.68	43.28	22.00	30.08	0.00	2.52	71.28
CWE	-	4.48	0.02	17.46	26.72	11.78	11.38	22.70	13.26	3.68	7.34
FWE	-	72.67	2.93	46.73	31.00	64.53	15.13	34.67	13.93	72.47	51.53
QA_1	-	20.20	5.20	20.60	33.40	34.40	23.00	28.00	6.00	22.60	27.00
QA_2	-	25.20	1.20	24.00	30.60	34.80	24.00	30.60	12.60	24.60	33.20
Avg.	-	12.34	0.72	29.50	57.66	59.42	36.95	52.17	3.53	13.56	60.03

表14：16k上下文长度上的RULER结果。

精确	冻结				微调						
	Base	Inf	N-F	SE	PI	N-32	YaRN	CLEX	LLR	Land	N-64
	✓		✓		✓	✓	✓	✓			✓
训练长度	4k	4k	4k	4k	32k	32k	32k	32k	32k	32k	64k
评估长度	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k
NIAH_S1	-	21.00	14.20	99.80	97.20	99.40	100.00	99.80	0.00	42.40	99.80
NIAH_S2	-	17.00	17.40	93.40	100.00	100.00	99.20	100.00	0.20	6.80	100.00
NIAH_S3	-	11.60	8.20	77.00	99.60	98.60	89.60	99.60	0.00	3.60	100.00
NIAH_M1	-	15.80	9.20	60.00	97.80	93.20	83.40	89.40	0.00	5.60	90.80
NIAH_M2	-	0.00	0.60	3.80	82.80	79.80	19.60	72.00	0.00	0.80	67.60
NIAH_M3	-	1.00	0.00	1.80	34.20	18.20	7.40	15.00	0.00	0.00	29.60
NIAH_MV	-	8.40	6.90	38.85	77.55	81.95	58.75	62.40	0.00	4.80	83.50
NIAH_MQ	-	8.85	7.95	59.30	90.95	86.20	85.15	81.60	0.00	2.75	90.35
VT	-	6.56	11.28	1.16	68.84	83.56	47.12	48.16	0.00	2.52	88.68
CWE	-	19.94	28.36	17.80	27.26	26.32	23.72	28.60	0.62	11.90	21.20
FWE	-	77.13	25.80	59.80	47.93	61.73	10.13	57.33	12.93	81.60	51.73
QA_1	-	22.80	36.40	28.00	46.00	45.20	43.20	49.20	13.20	23.00	45.00
QA_2	-	24.20	26.00	31.60	35.20	36.00	37.40	33.40	20.80	26.20	36.00
Avg.	-	18.02	14.79	44.02	69.64	70.01	54.21	64.35	3.67	16.31	69.56

表15：在32k上下文长度上的RULER结果。

精确	冻结				微调						
	Base	Inf	N-F	SE	PI	N-32	YaRN	CLEX	LLR	Land	N-64
	✓		✓		✓	✓	✓	✓			✓
训练长度	4k	4k	4k	4k	32k	32k	32k	32k	32k	32k	64k
评估长度	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k
NIAH S1_	-	7.80	0.00	83.00	97.20	99.00	85.80	85.60	0.00	33.80	100.00
NIAH S2_	-	7.00	0.00	68.40	99.00	100.00	81.00	94.00	0.00	3.20	99.20
NIAH S3_	-	6.40	0.00	42.80	97.00	99.40	62.40	97.20	0.00	2.60	96.40
NIAH M1_	-	8.80	0.00	29.40	93.40	90.80	63.20	78.40	0.00	5.40	82.60
NIAH M2_	-	0.00	0.00	2.40	48.80	39.40	6.40	40.40	0.00	0.20	36.60
NIAH_M3	-	0.00	0.00	1.40	5.80	8.60	1.20	8.00	0.00	0.00	7.20
NIAH_MV	-	3.75	0.00	24.65	61.30	68.20	37.95	60.75	0.00	2.80	82.20
NIAH_MQ	-	2.05	0.00	20.35	68.65	78.25	46.95	67.80	0.05	2.35	85.80
VT	-	2.08	0.00	2.32	56.68	43.28	22.00	30.08	0.00	2.52	71.28
CWE	-	4.48	0.02	17.46	26.72	11.78	11.38	22.70	13.26	3.68	7.34
FWE	-	72.67	2.93	46.73	31.00	64.53	15.13	34.67	13.93	72.47	51.53
QA_1	-	20.20	5.20	20.60	33.40	34.40	23.00	28.00	6.00	22.60	27.00
QA_2	-	25.20	1.20	24.00	30.60	34.80	24.00	30.60	12.60	24.60	33.20
Avg.	-	12.34	0.72	29.50	57.66	59.42	36.95	52.17	3.53	13.56	60.03

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Table 16: Ruler results on 64k context length.

Exact	Base	Inf	N-F	SE	PI	N-32	YaRN	CLEX	LLR	Land	N-64
	Frozen				Finetuned						
	✓		✓		✓	✓	✓	✓			✓
Train Len	4k	4k	4k	4k	32k	32k	32k	32k	32k	32k	64k
Eval Len	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k
NIAH_S1	-	3.20	0.00	71.80	0.00	83.60	0.00	40.60	0.00	40.00	98.00
NIAH_S2	-	3.80	0.00	0.20	0.00	95.60	0.00	68.80	0.00	3.00	98.00
NIAH_S3	-	5.40	0.00	0.00	0.00	95.40	0.00	70.40	0.00	3.00	95.80
NIAH_M1	-	5.40	0.00	0.00	0.00	76.80	0.00	55.40	0.00	5.20	67.20
NIAH_M2	-	0.00	0.00	2.60	0.00	15.20	0.00	15.80	0.00	0.00	25.80
NIAH_M3	-	0.00	0.00	0.20	0.00	1.20	0.00	1.00	0.00	0.00	4.00
NIAH_MV	-	4.45	0.00	0.20	0.00	51.70	0.00	36.40	0.00	3.70	51.20
NIAH_MQ	-	4.45	0.00	0.05	0.00	56.60	0.00	43.50	0.00	2.45	65.40
VT	-	1.28	0.00	12.20	0.00	34.28	0.00	0.00	0.00	2.40	41.48
CWE	-	0.76	0.00	6.85	0.00	6.58	0.00	9.72	0.00	1.70	7.88
FWE	-	72.20	11.47	26.47	0.00	25.27	0.00	11.73	0.00	82.67	27.73
QA_1	-	16.20	0.20	0.80	0.00	30.80	0.00	25.60	0.00	19.60	29.20
QA_2	-	20.20	0.20	0.00	0.00	28.40	0.00	19.00	0.00	20.20	29.40
Avg.	-	10.56	0.91	9.34	0.00	46.26	0.00	30.61	0.00	14.15	49.31

表16：在64k上下文长度上的RULER结果。

精确	Base	Inf	N-F	SE	PI	N-32	YaRN	CLEX	LLR	Land	N-64
	冻结				微调						
	✓		✓		✓	✓	✓	✓			✓
训练长度	4k	4k	4k	4k	32k	32k	32k	32k	32k	32k	64k
评估长度	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k	4k
NIAH_S1	-	3.20	0.00	71.80	0.00	83.60	0.00	40.60	0.00	40.00	98.00
NIAH_S2	-	3.80	0.00	0.20	0.00	95.60	0.00	68.80	0.00	3.00	98.00
NIAH_S3	-	5.40	0.00	0.00	0.00	95.40	0.00	70.40	0.00	3.00	95.80
NIAH_M1	-	5.40	0.00	0.00	0.00	76.80	0.00	55.40	0.00	5.20	67.20
NIAH_M2	-	0.00	0.00	2.60	0.00	15.20	0.00	15.80	0.00	0.00	25.80
NIAH_M3	-	0.00	0.00	0.20	0.00	1.20	0.00	1.00	0.00	0.00	4.00
NIAH_MV	-	4.45	0.00	0.20	0.00	51.70	0.00	36.40	0.00	3.70	51.20
NIAH_MQ	-	4.45	0.00	0.05	0.00	56.60	0.00	43.50	0.00	2.45	65.40
VT	-	1.28	0.00	12.20	0.00	34.28	0.00	0.00	0.00	2.40	41.48
CWE	-	0.76	0.00	6.85	0.00	6.58	0.00	9.72	0.00	1.70	7.88
FWE	-	72.20	11.47	26.47	0.00	25.27	0.00	11.73	0.00	82.67	27.73
QA_1	-	16.20	0.20	0.80	0.00	30.80	0.00	25.60	0.00	19.60	29.20
QA_2	-	20.20	0.20	0.00	0.00	28.40	0.00	19.00	0.00	20.20	29.40
Avg.	-	10.56	0.91	9.34	0.00	46.26	0.00	30.61	0.00	14.15	49.31